

BFS Traversal

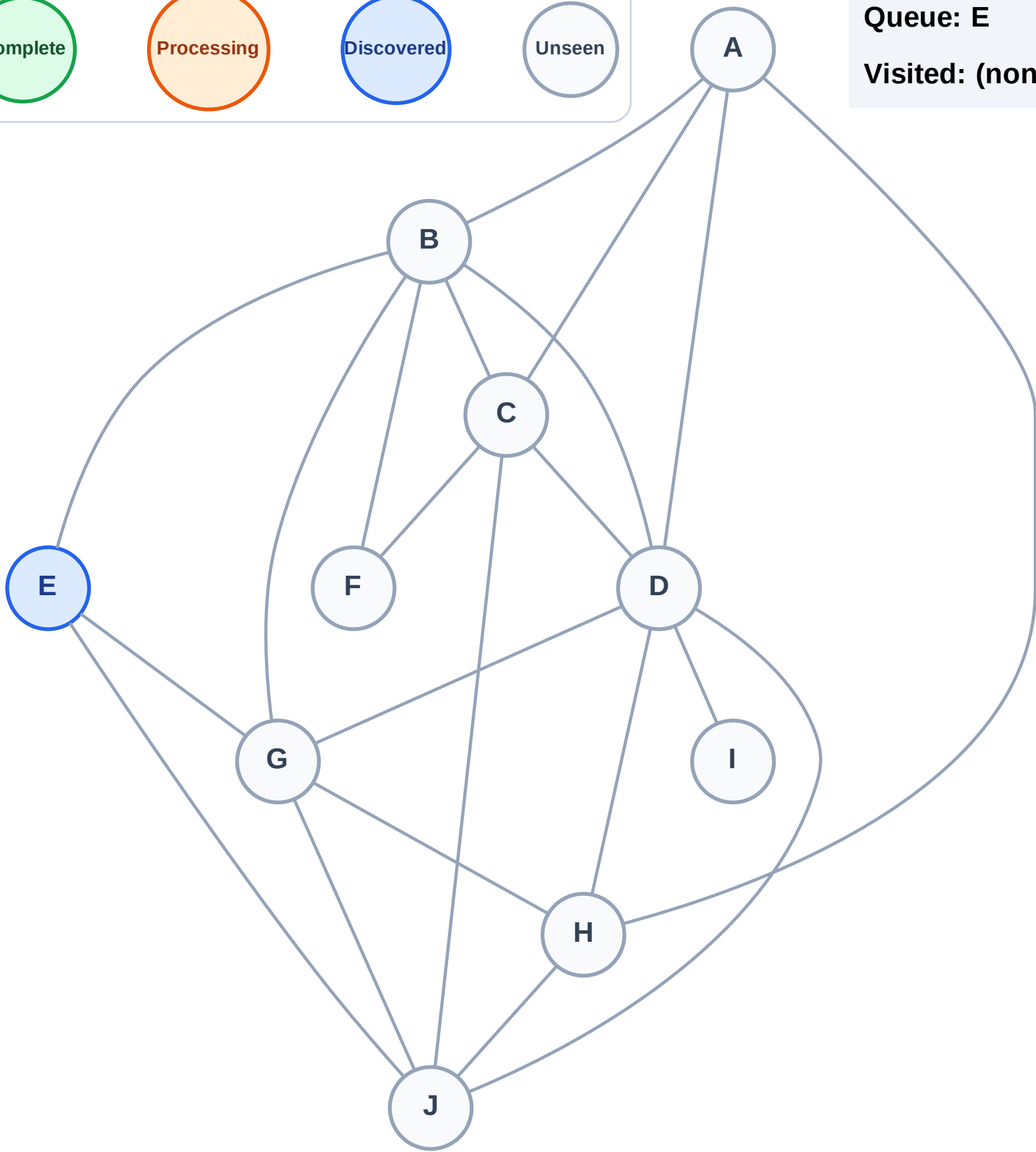
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

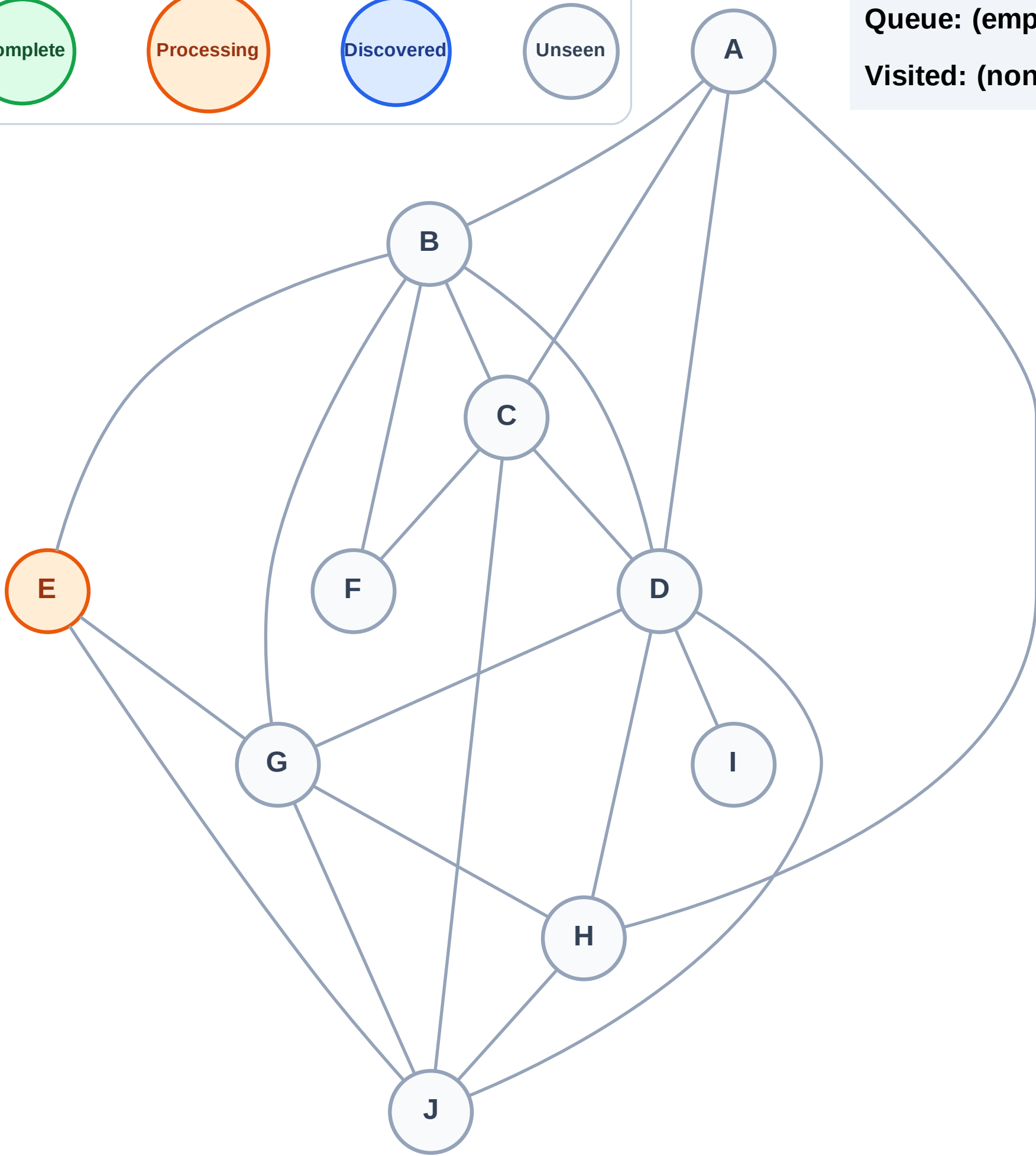
Step 2: Process node E

Legend



Queue: (empty)

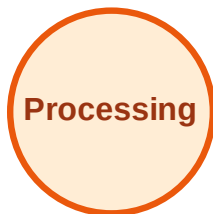
Visited: (none)



BFS Traversal

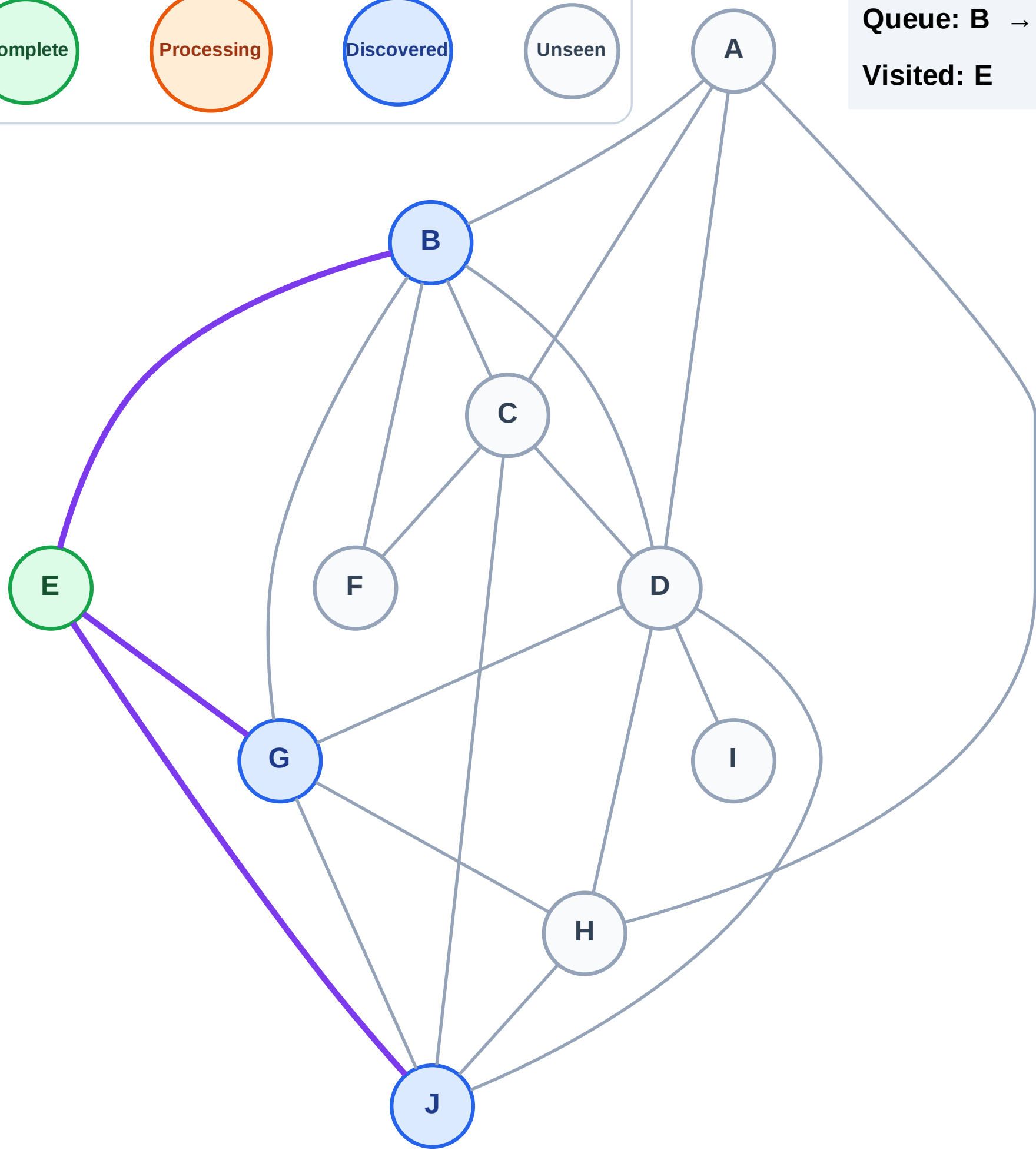
Step 3: Discover B, G, J from E

Legend



Queue: B → G → J

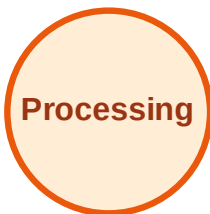
Visited: E



BFS Traversal

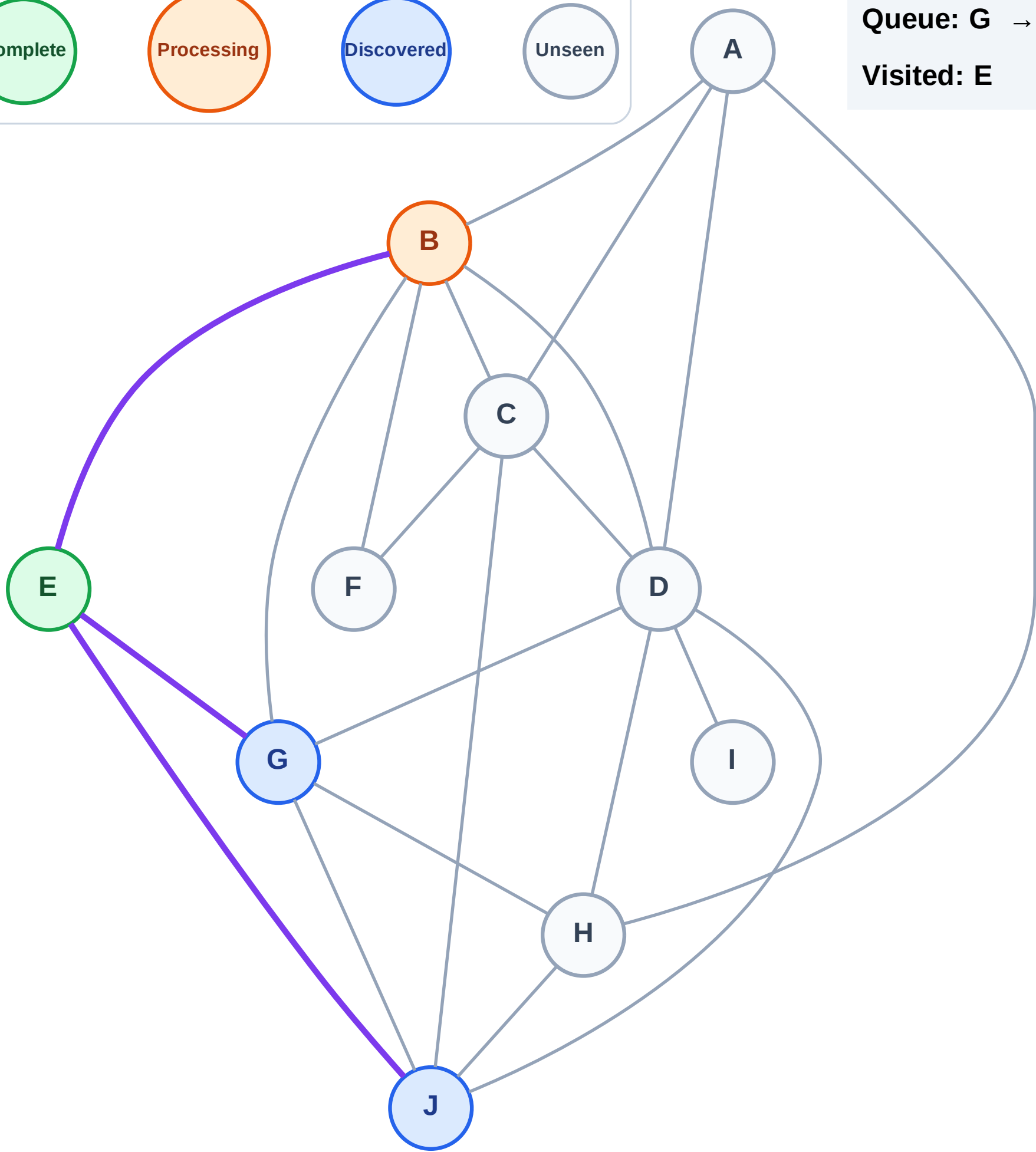
Step 4: Process node B

Legend



Queue: G → J

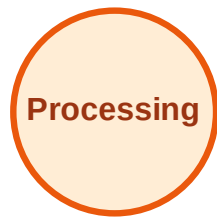
Visited: E



BFS Traversal

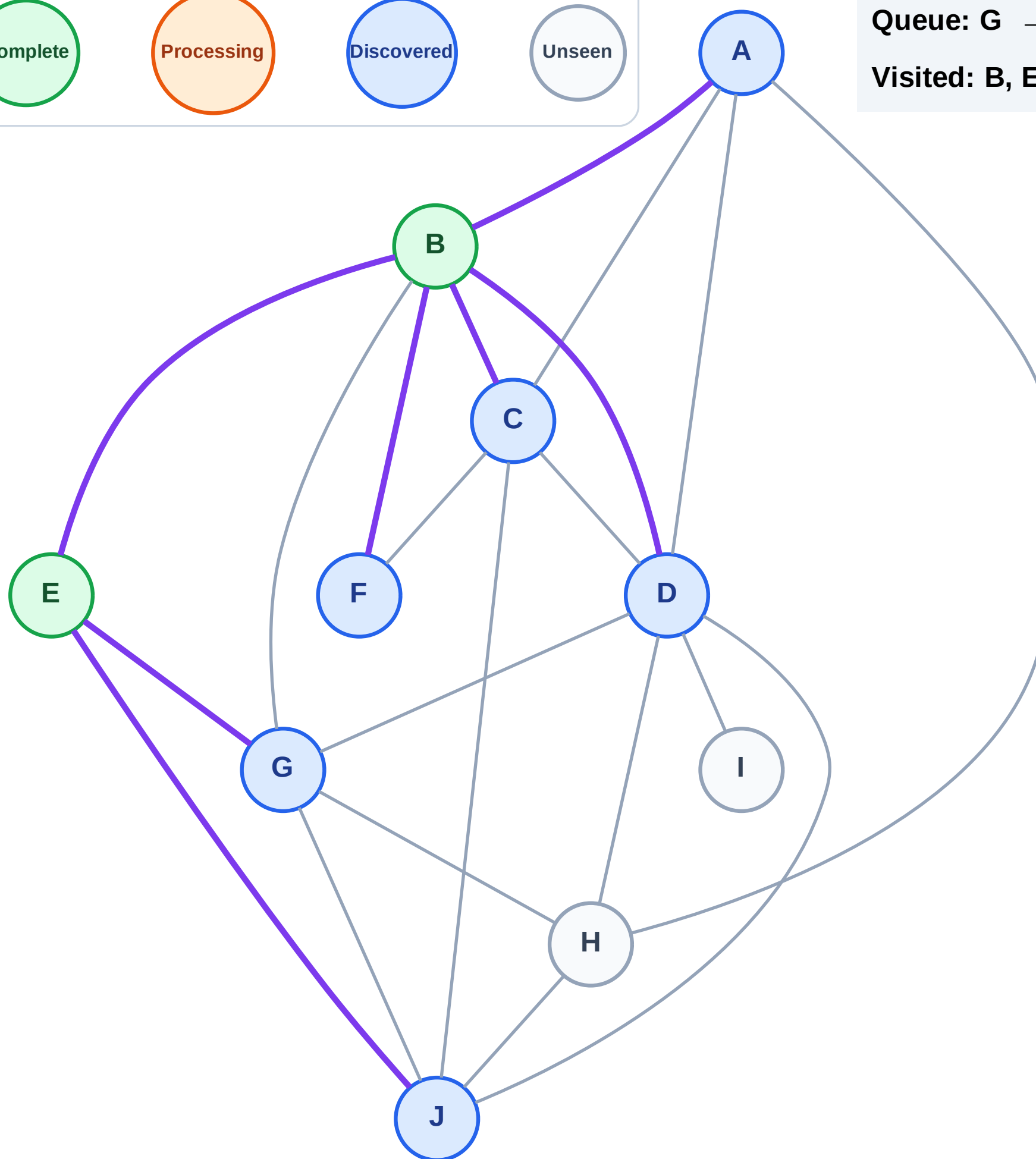
Step 5: Discover A, C, D, F from B

Legend



Queue: G → J → A → C → D → F

Visited: B, E



BFS Traversal

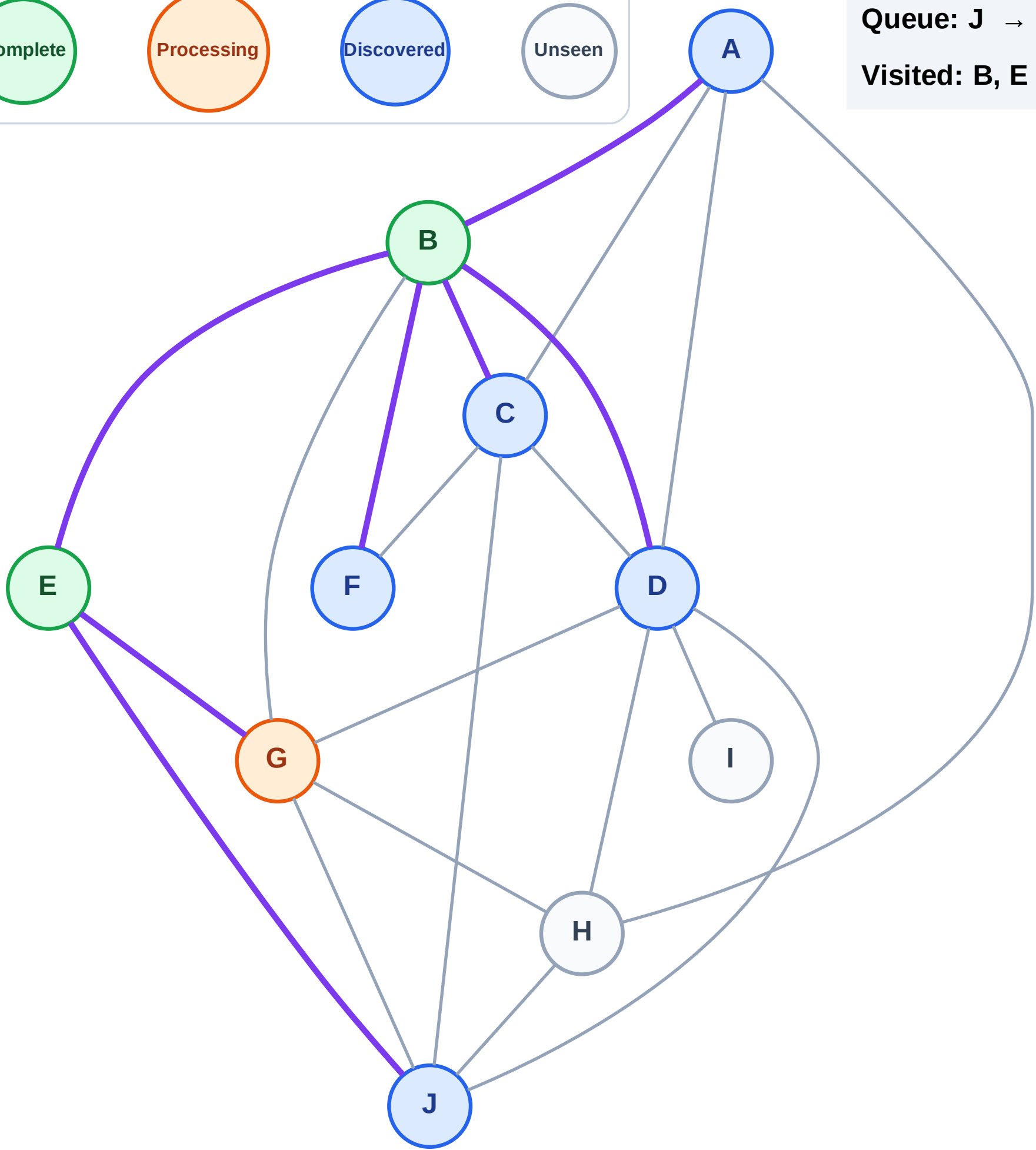
Step 6: Process node G

Legend



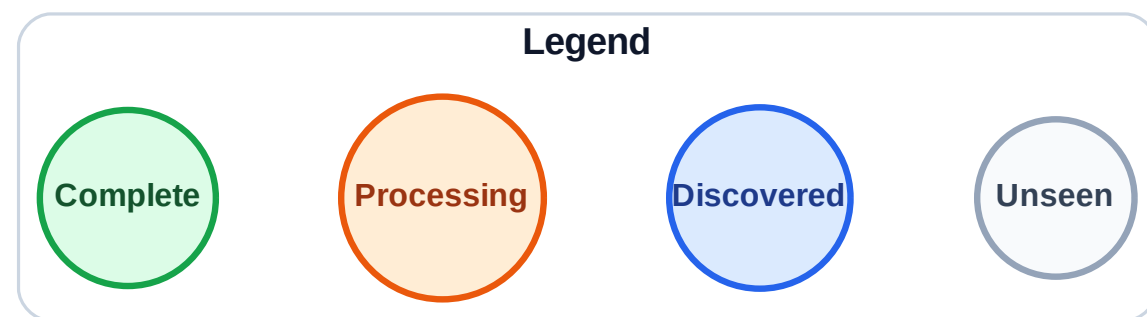
Queue: J → A → C → D → F

Visited: B, E

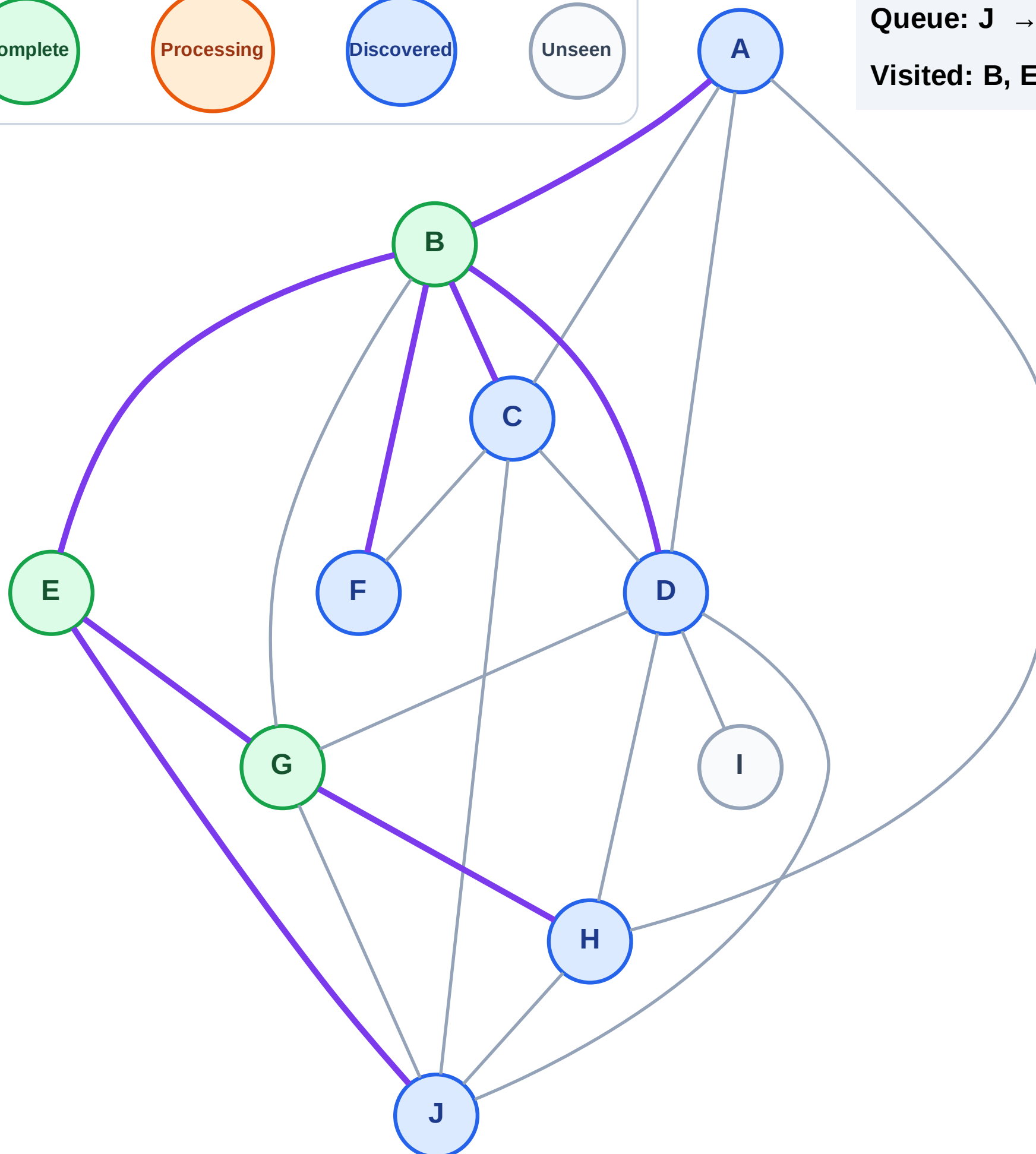


BFS Traversal

Step 7: Discover H from G



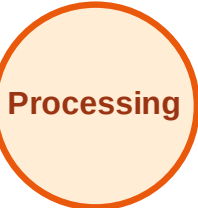
Queue: J → A → C → D → F → H
Visited: B, E, G



BFS Traversal

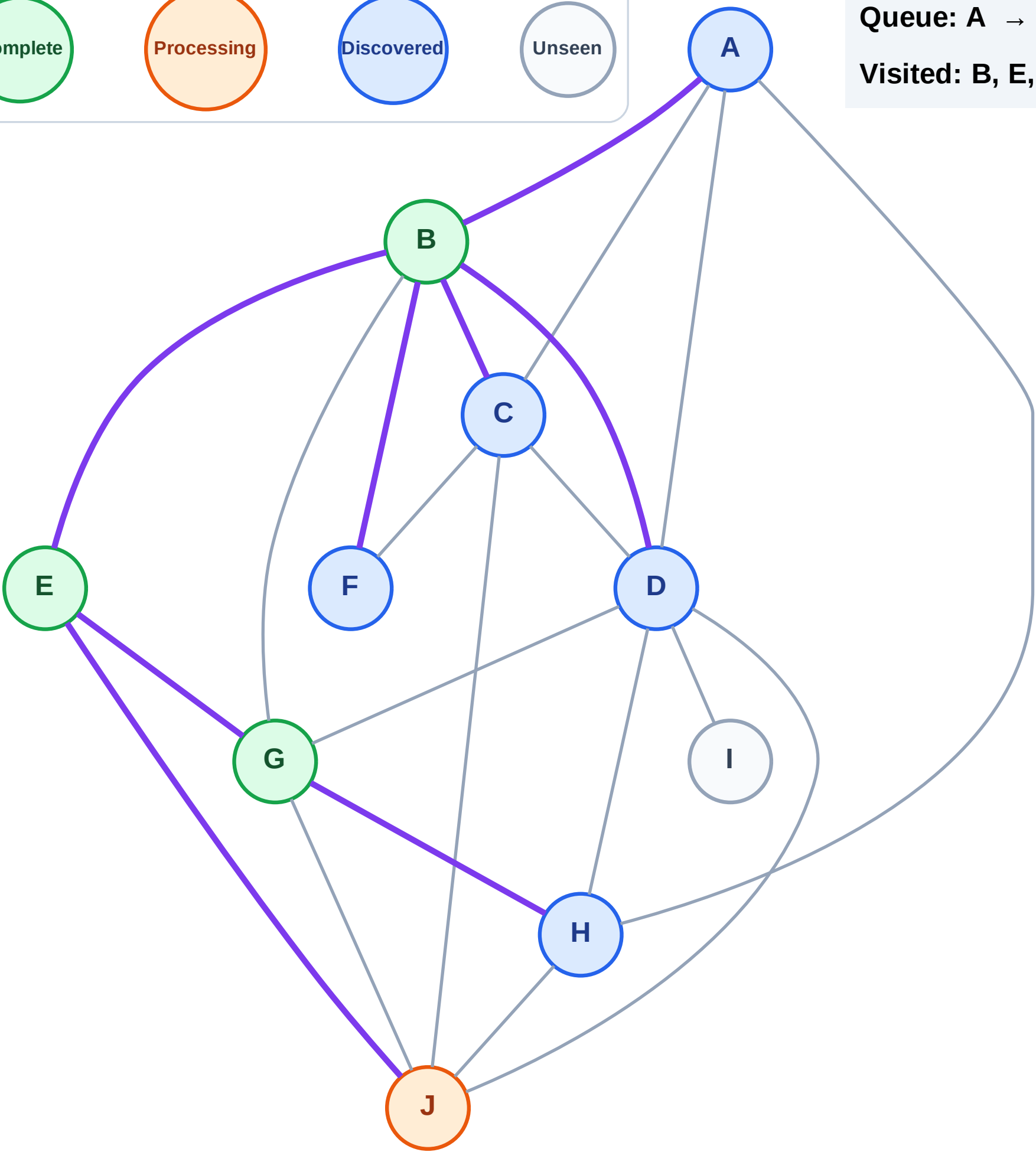
Step 8: Process node J

Legend



Queue: A → C → D → F → H

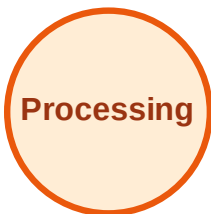
Visited: B, E, G



BFS Traversal

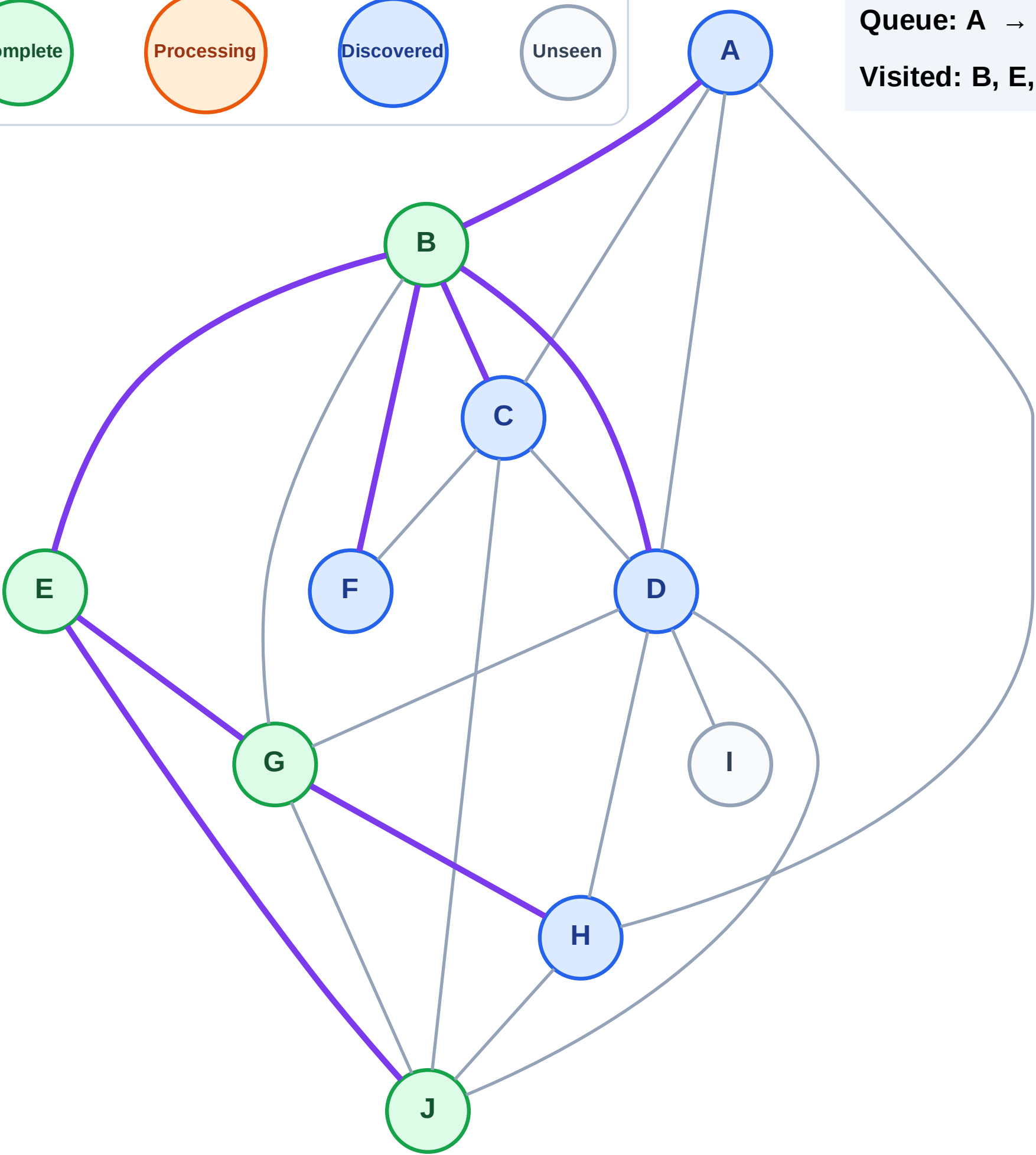
Step 9: No new neighbors from J

Legend



Queue: A → C → D → F → H

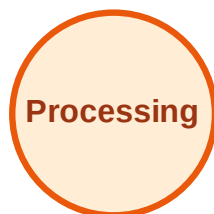
Visited: B, E, G, J



BFS Traversal

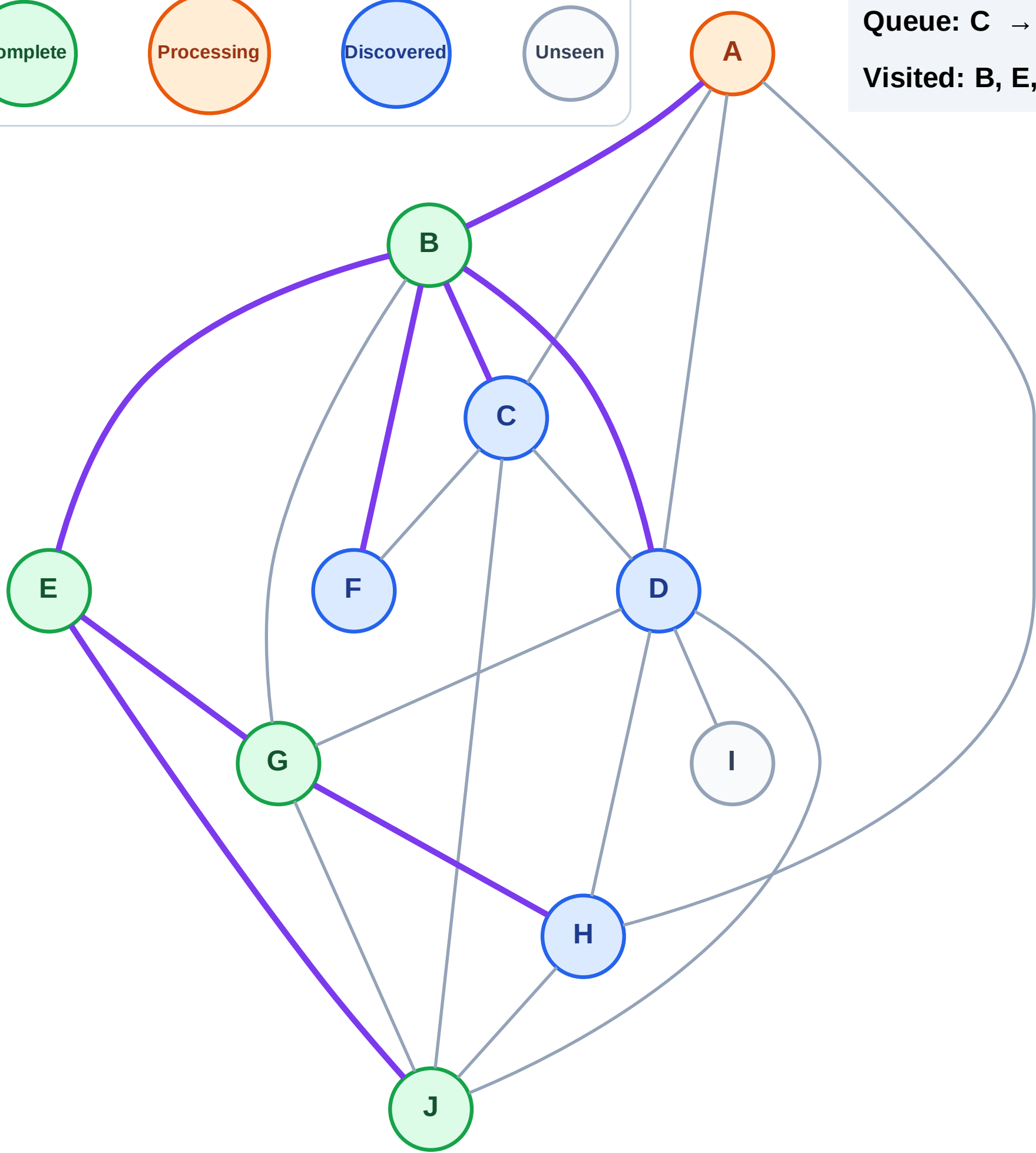
Step 10: Process node A

Legend



Queue: C → D → F → H

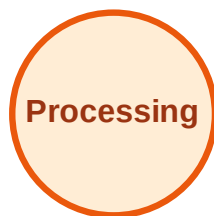
Visited: B, E, G, J



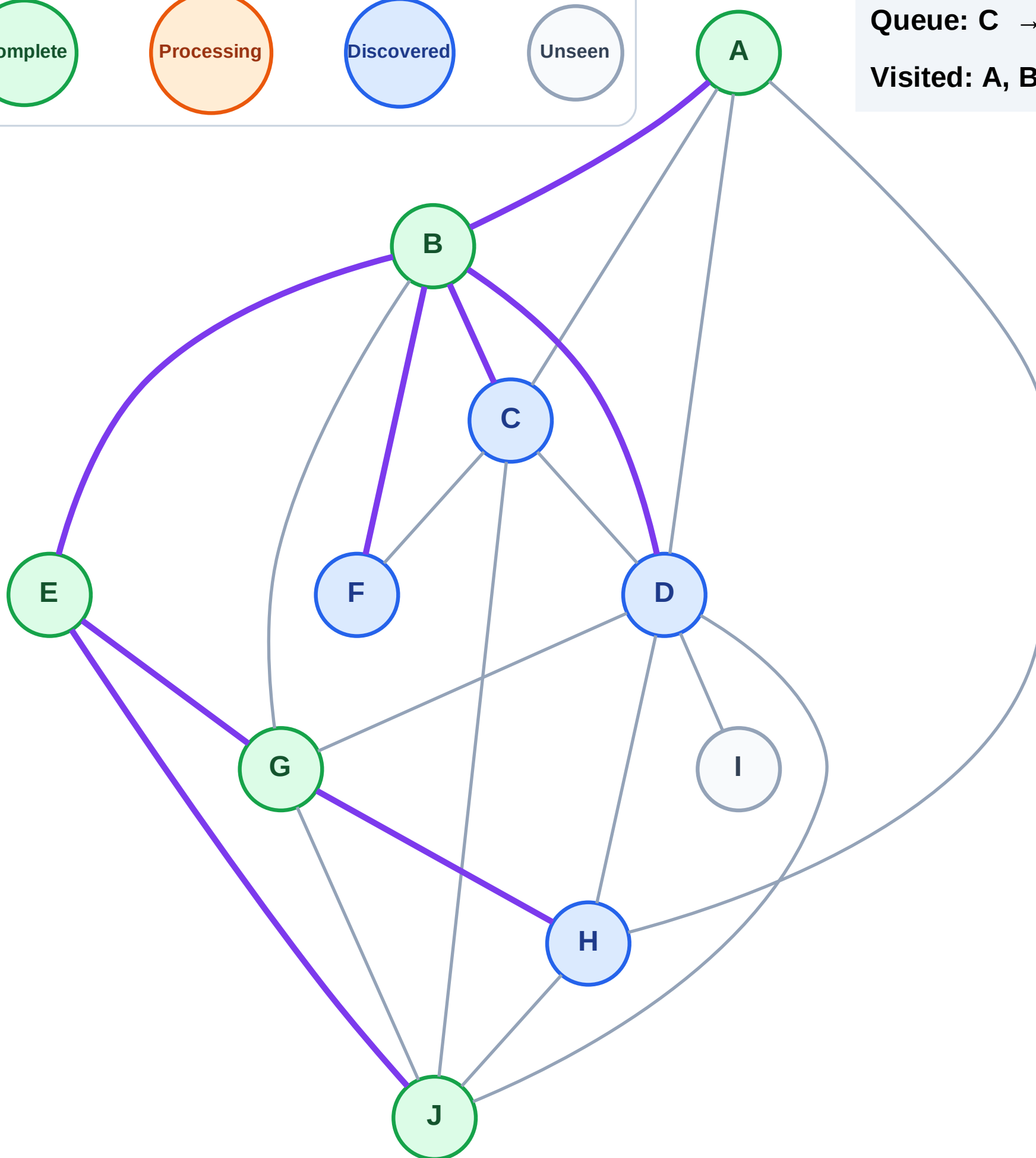
BFS Traversal

Step 11: No new neighbors from A

Legend



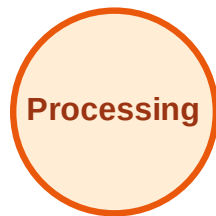
Queue: C → D → F → H
Visited: A, B, E, G, J



BFS Traversal

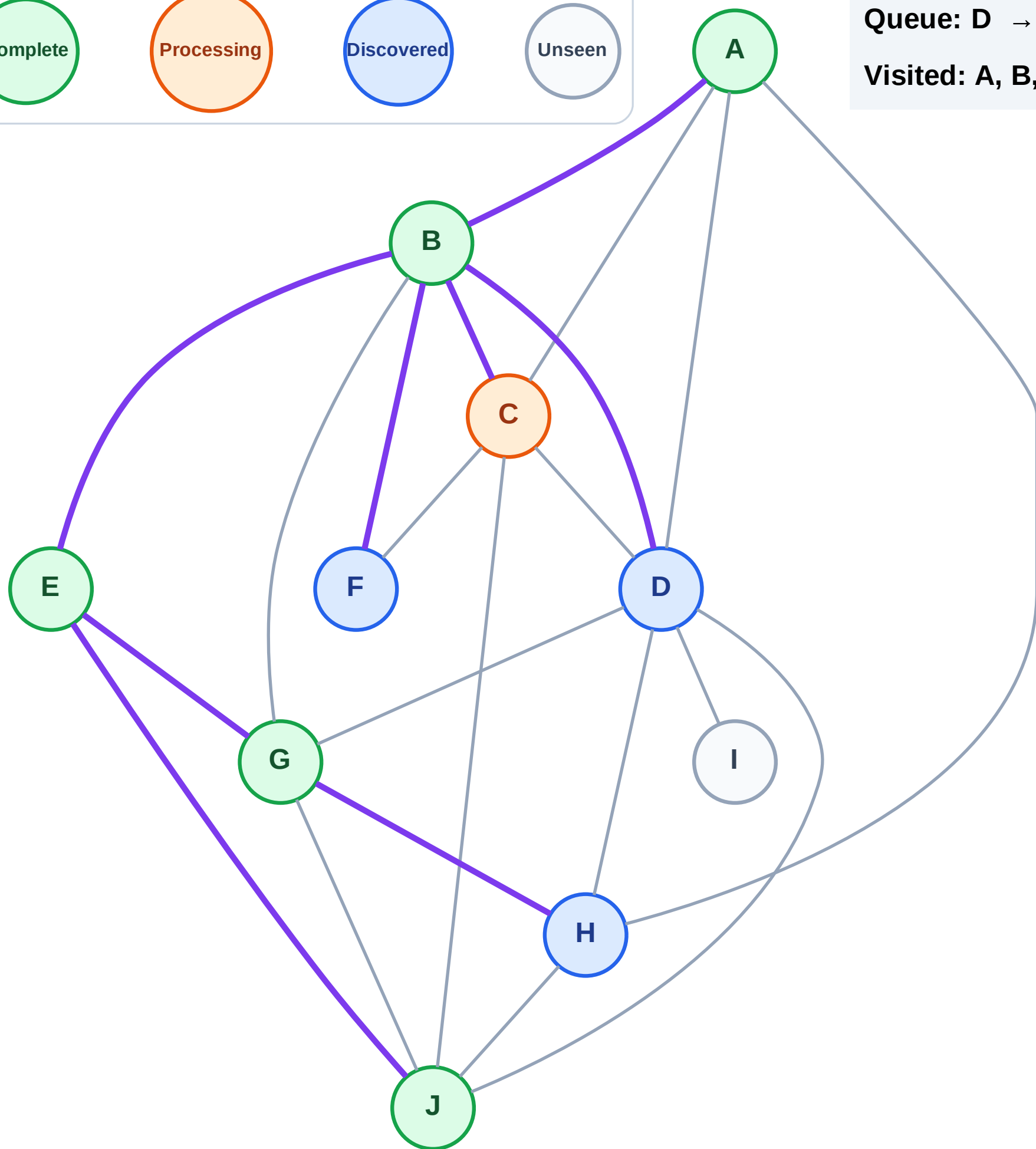
Step 12: Process node C

Legend



Queue: D → F → H

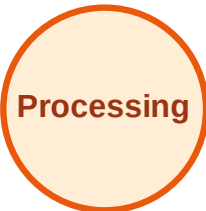
Visited: A, B, E, G, J



BFS Traversal

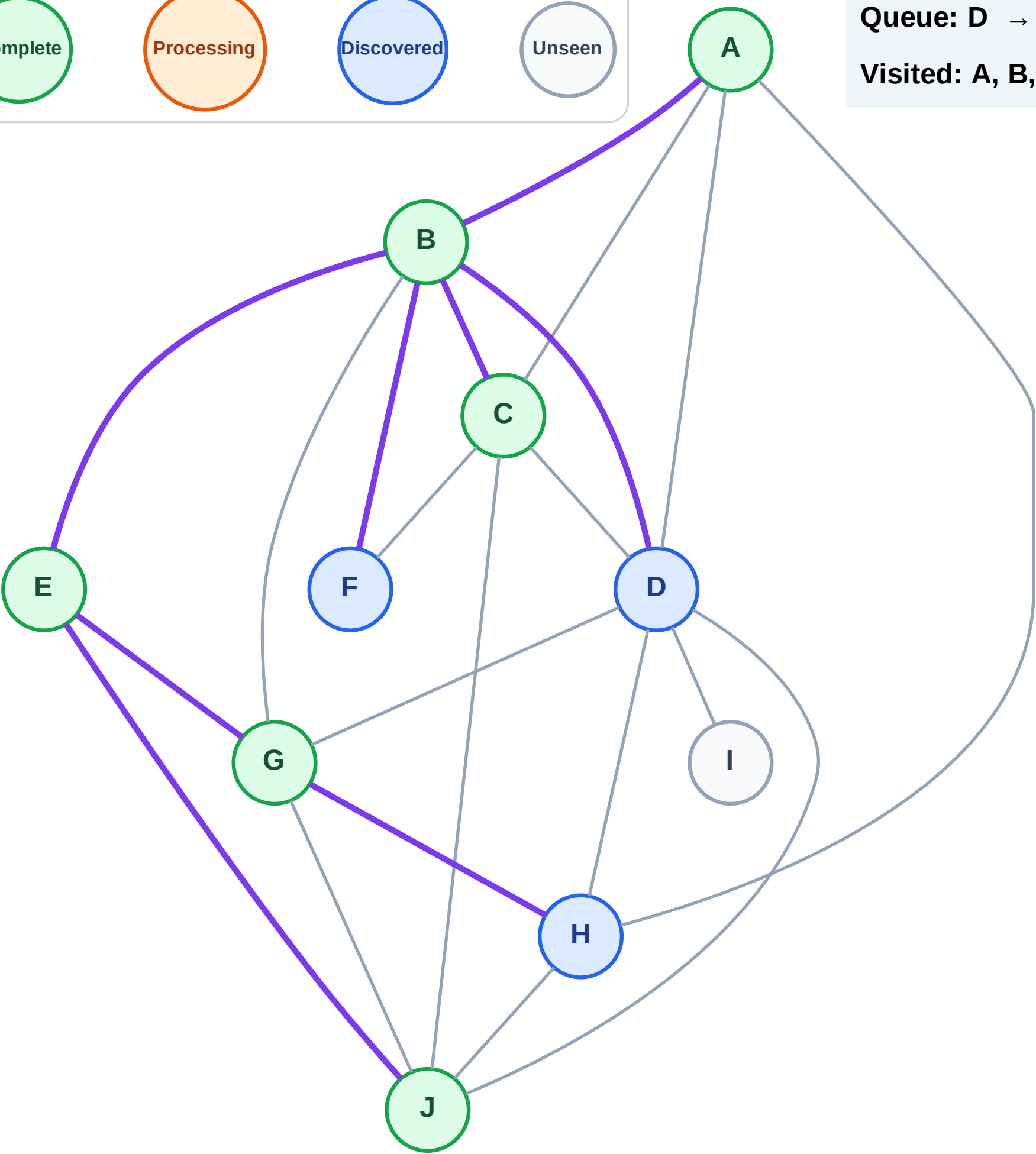
Step 13: No new neighbors from C

Legend



Queue: D → F → H

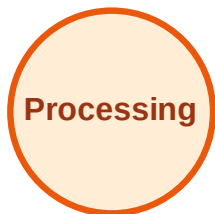
Visited: A, B, C, E, G, J



BFS Traversal

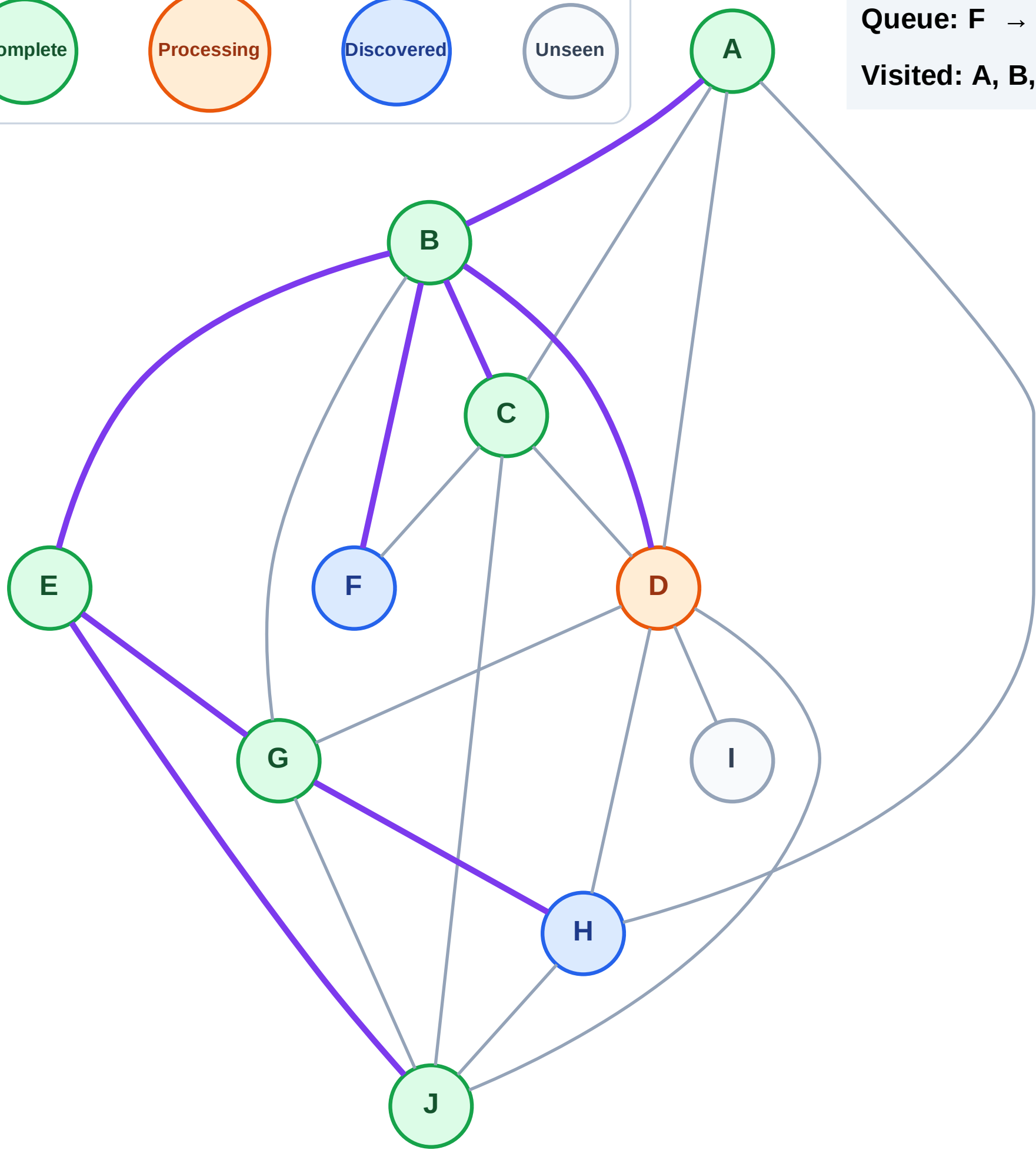
Step 14: Process node D

Legend



Queue: F → H

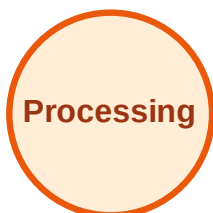
Visited: A, B, C, E, G, J



BFS Traversal

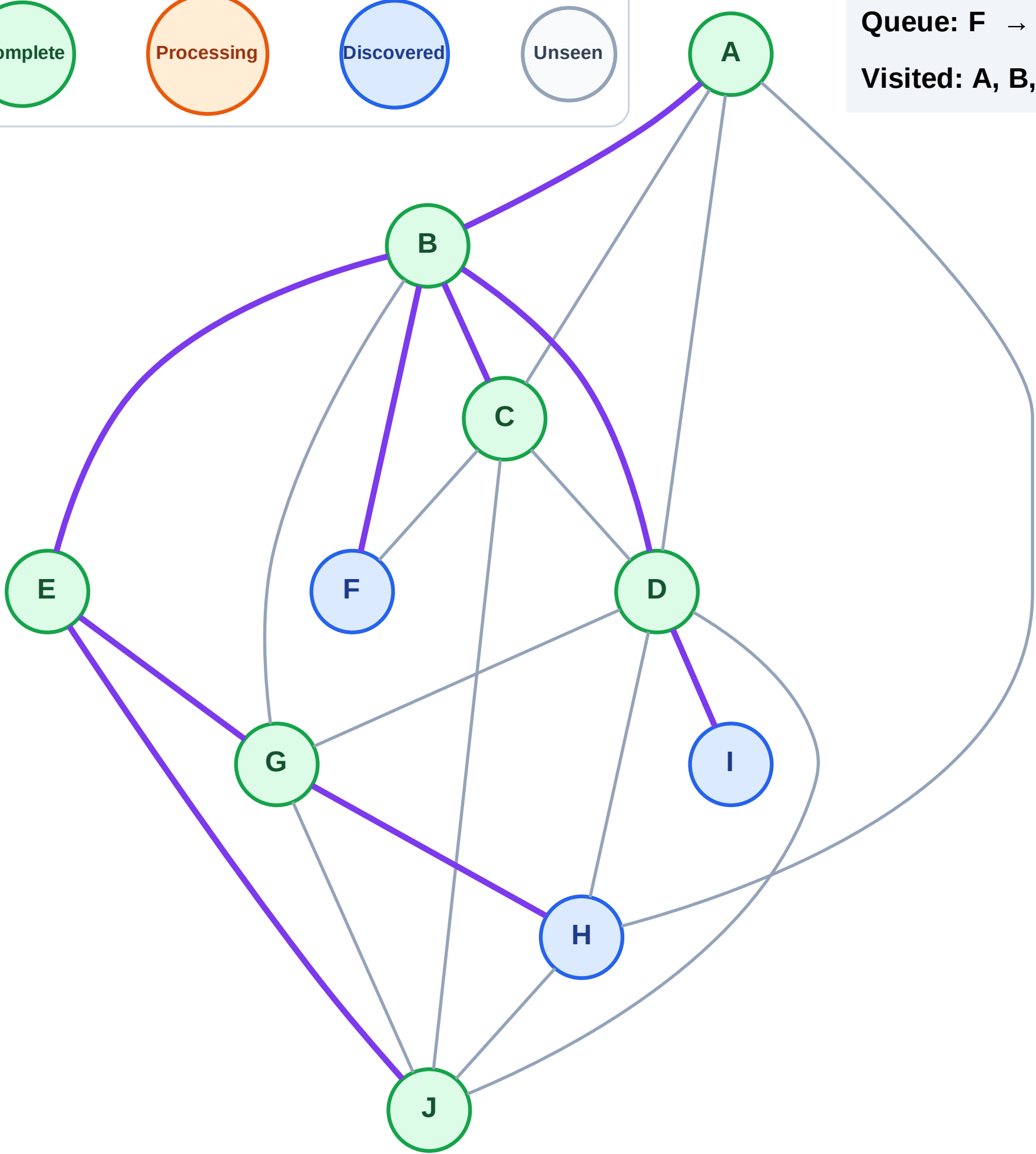
Step 15: Discover I from D

Legend



Queue: F → H → I

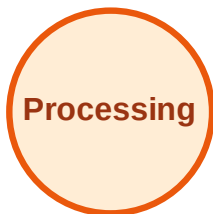
Visited: A, B, C, D, E, G, J



BFS Traversal

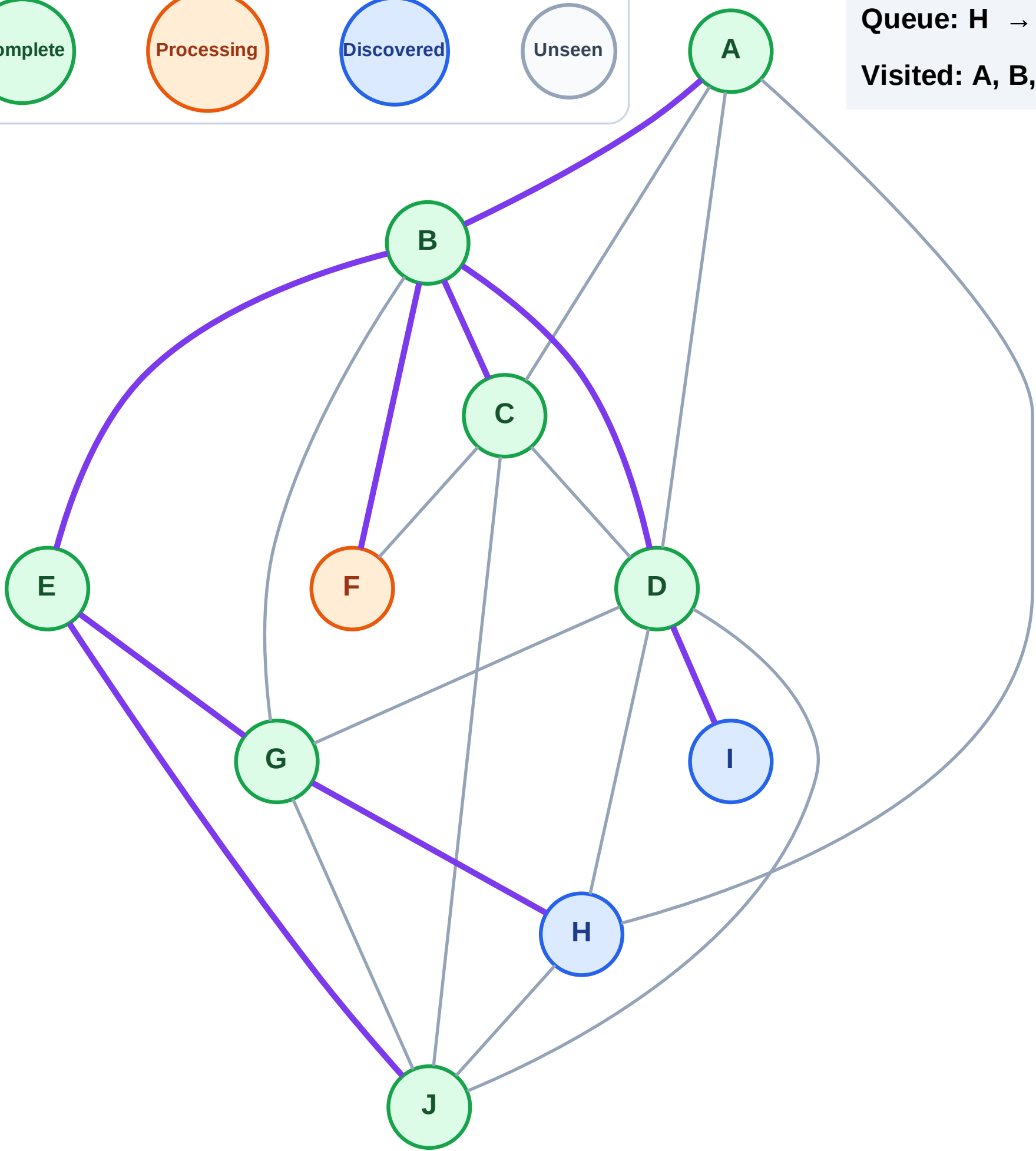
Step 16: Process node F

Legend



Queue: H → I

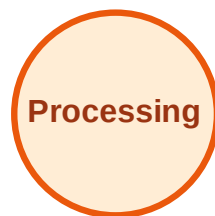
Visited: A, B, C, D, E, G, J



BFS Traversal

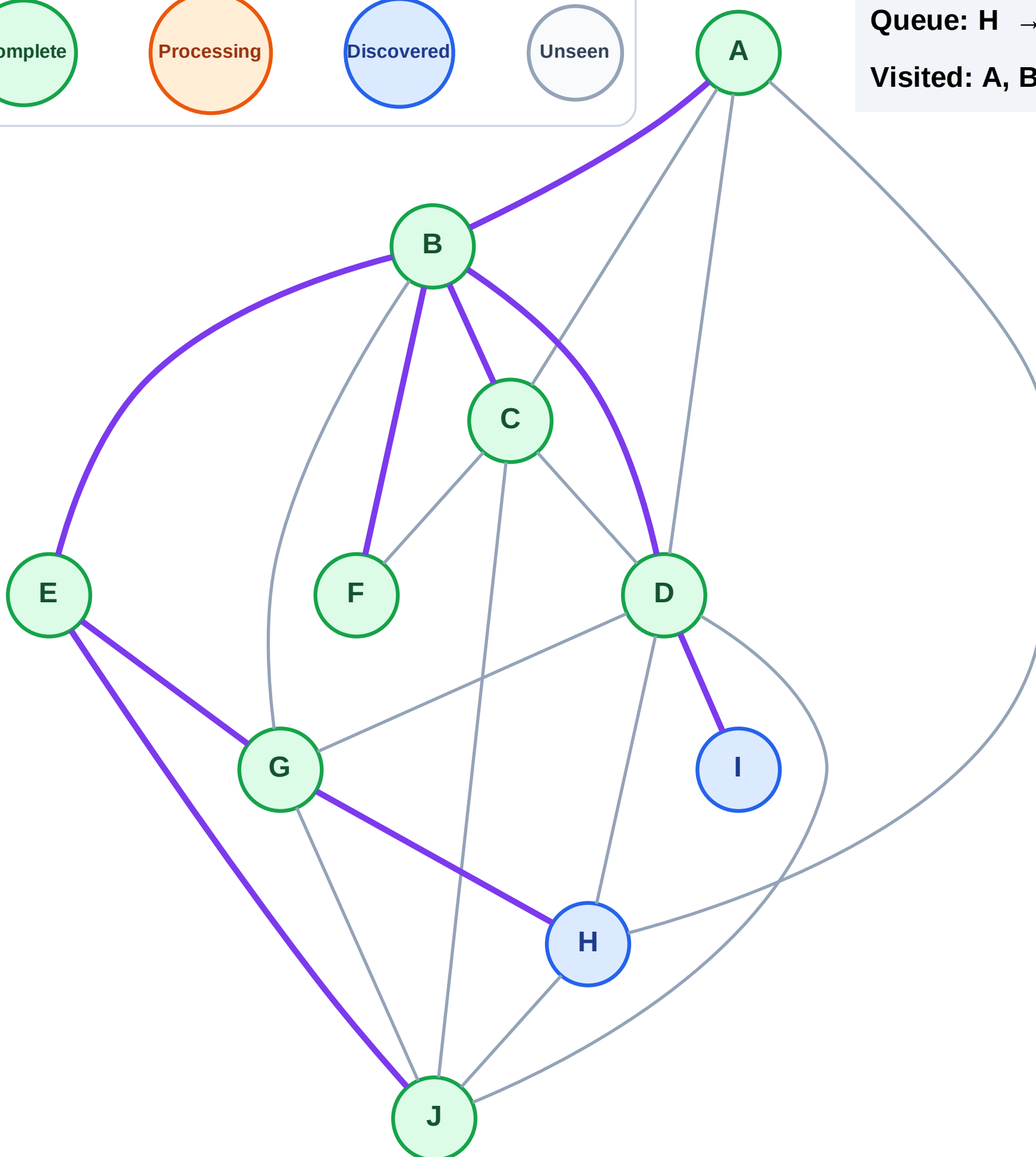
Step 17: No new neighbors from F

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node H

Legend

Complete

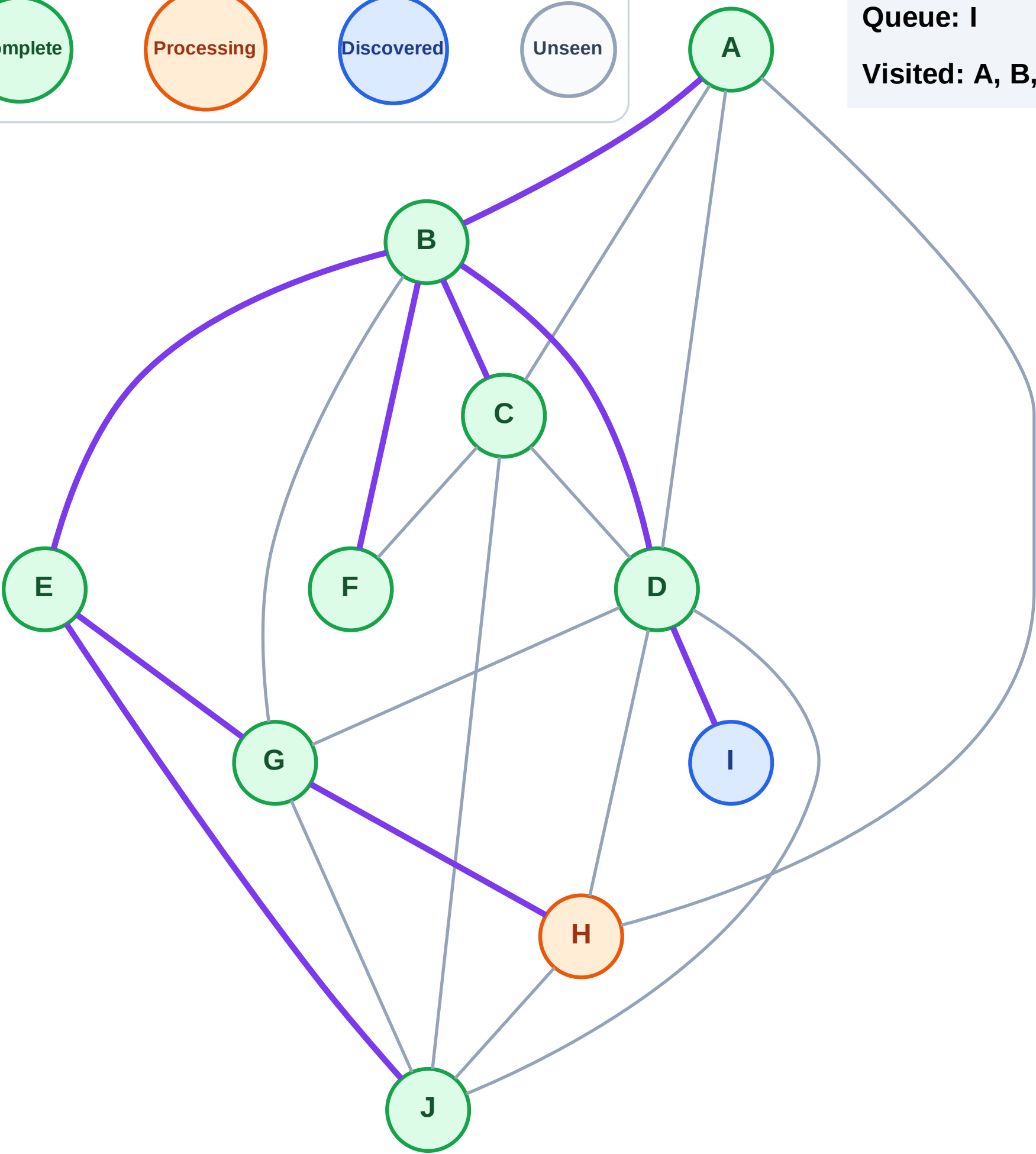
Processing

Discovered

Unseen

Queue: I

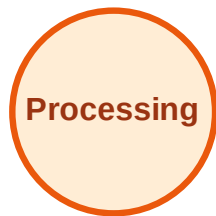
Visited: A, B, C, D, E, F, G, J



BFS Traversal

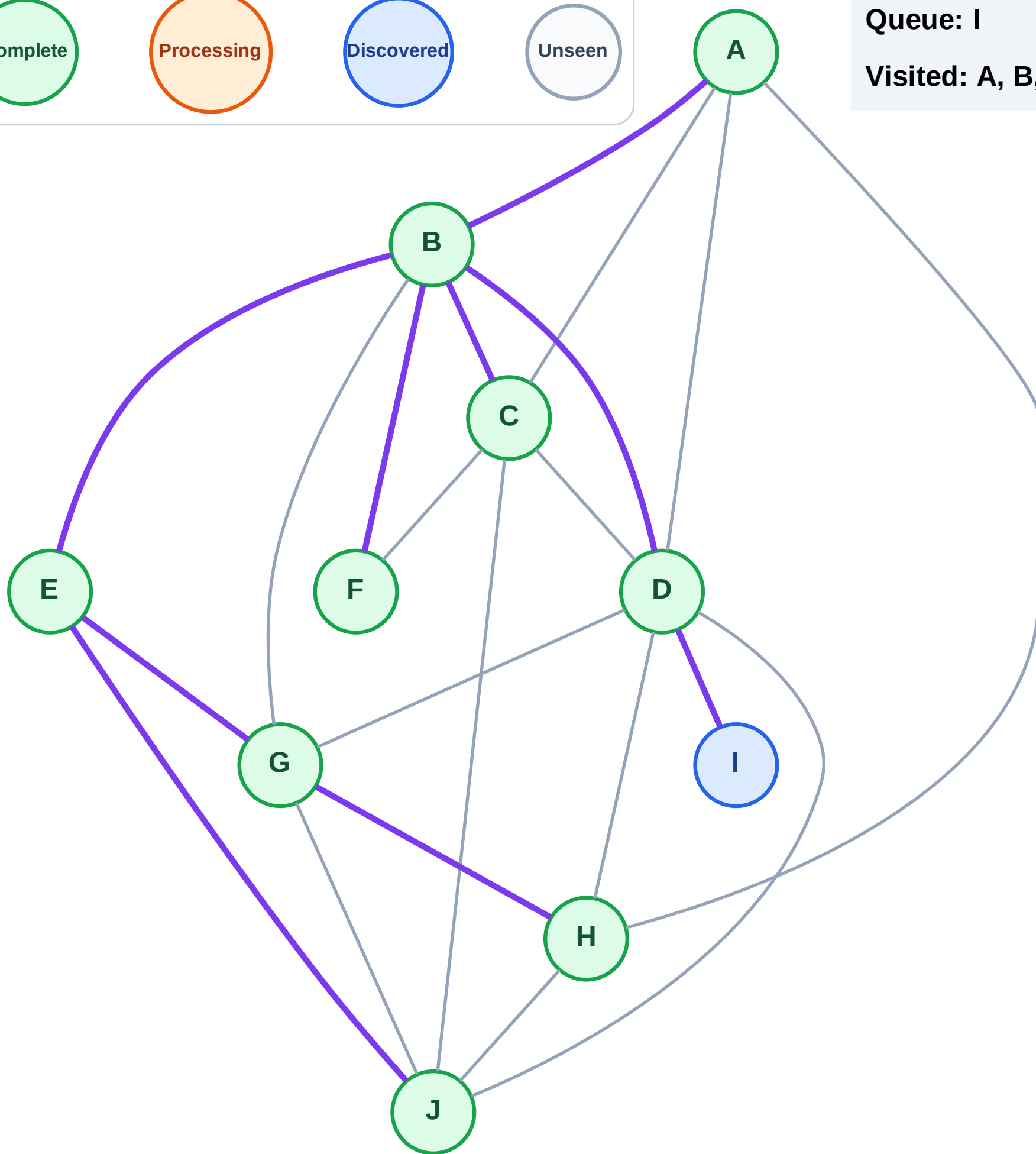
Step 19: No new neighbors from H

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

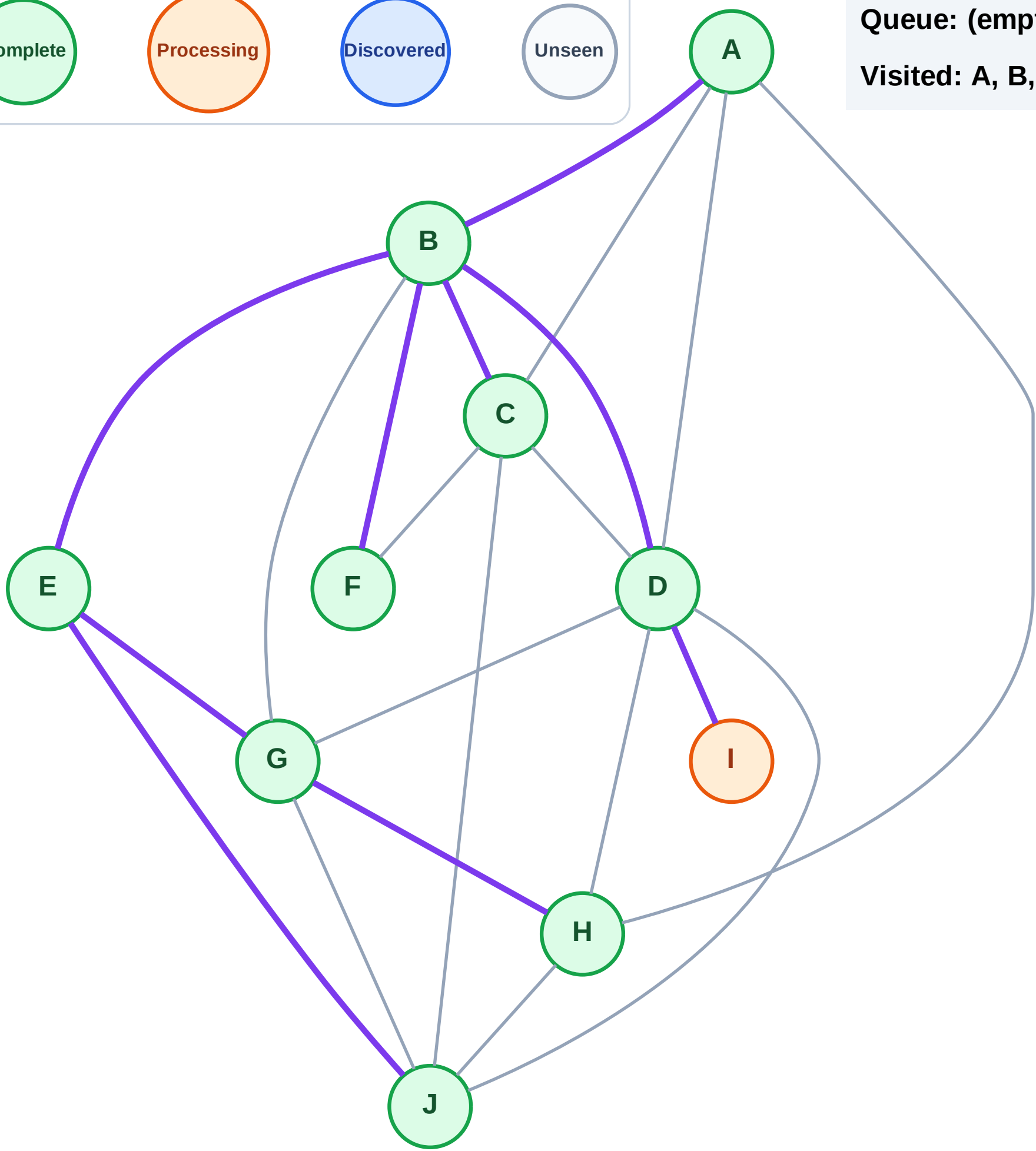
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

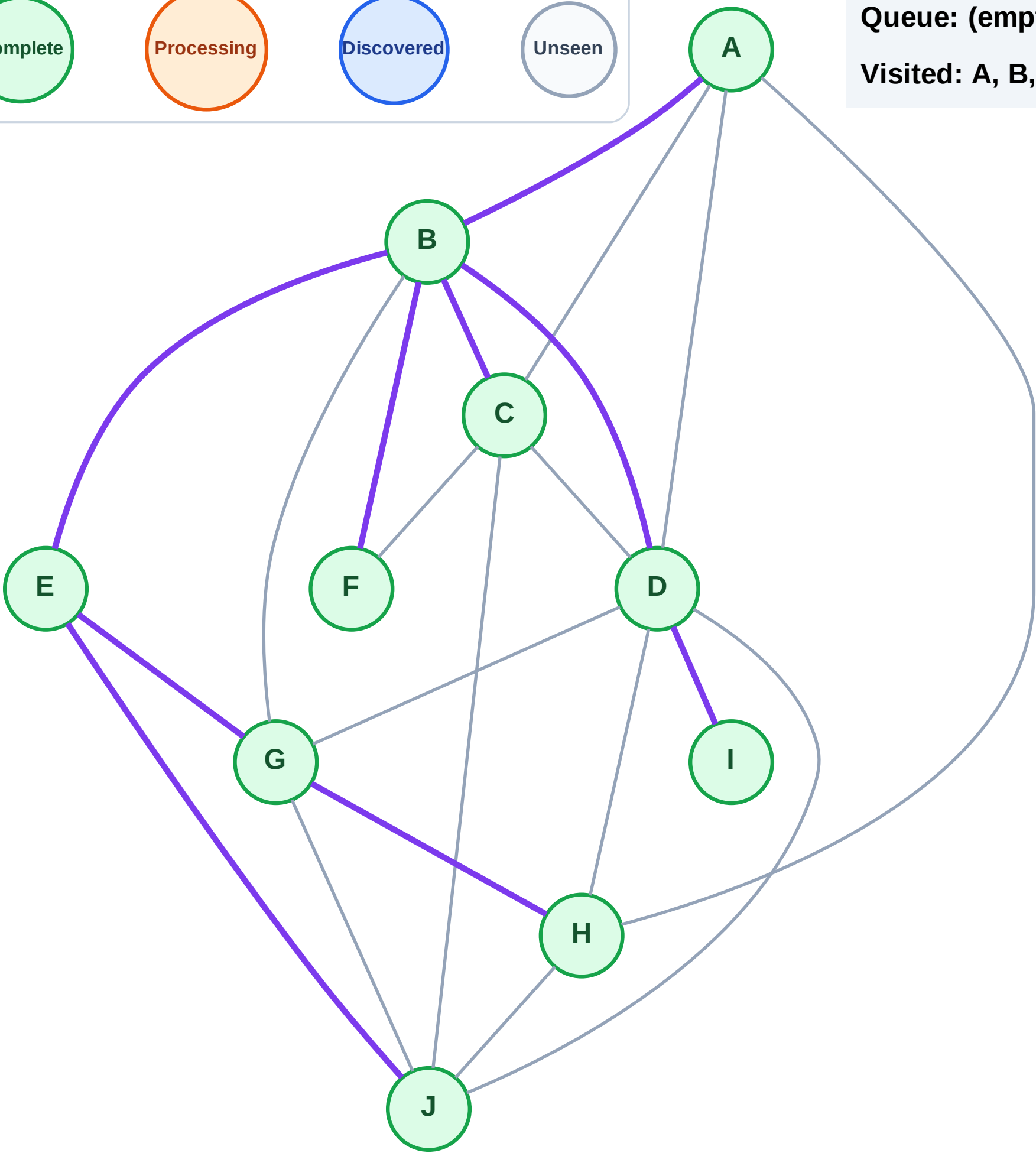
Step 21: No new neighbors from I

Legend



Queue: (empty)

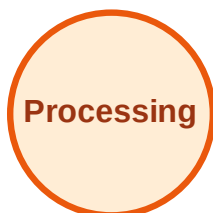
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

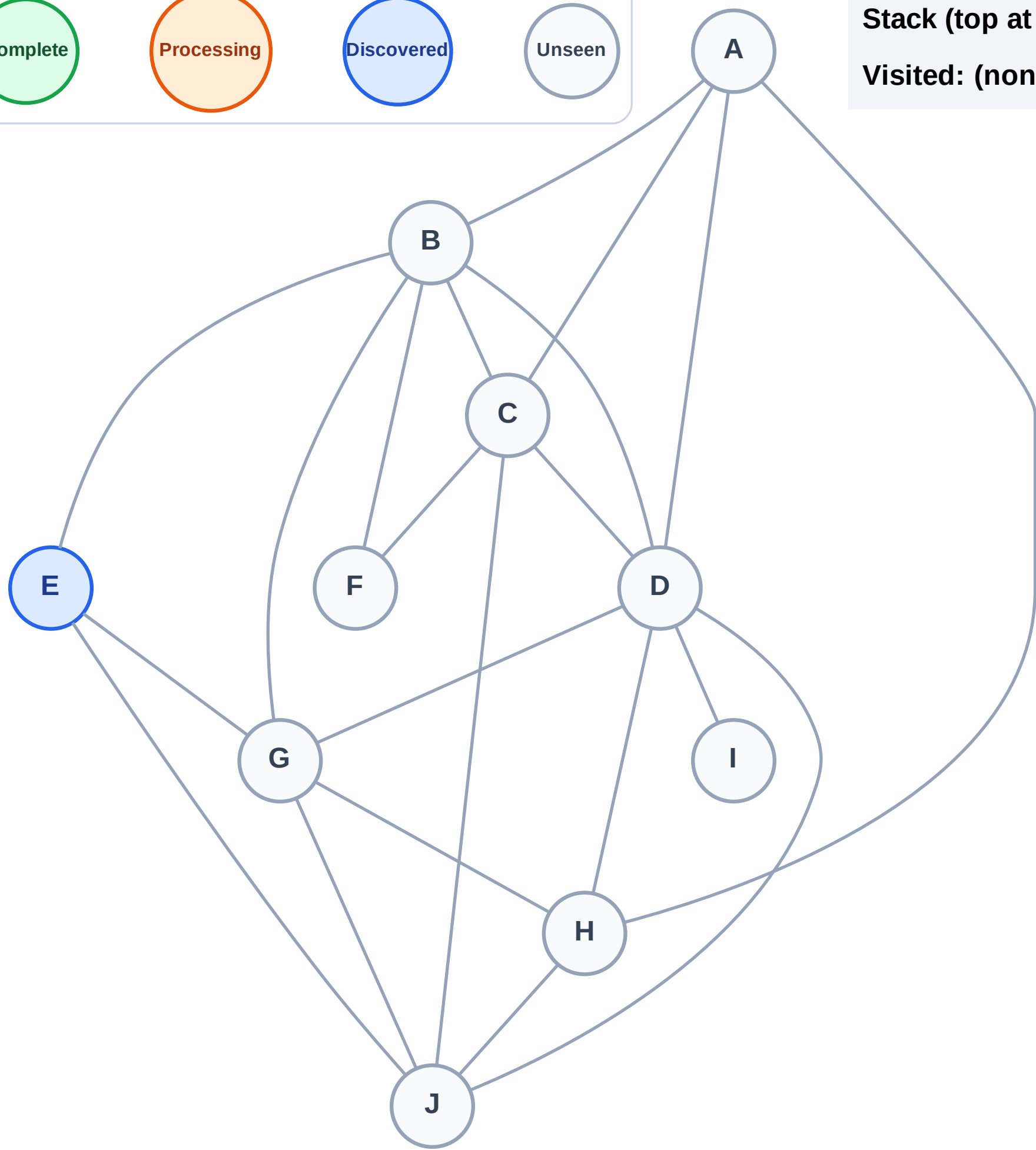
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

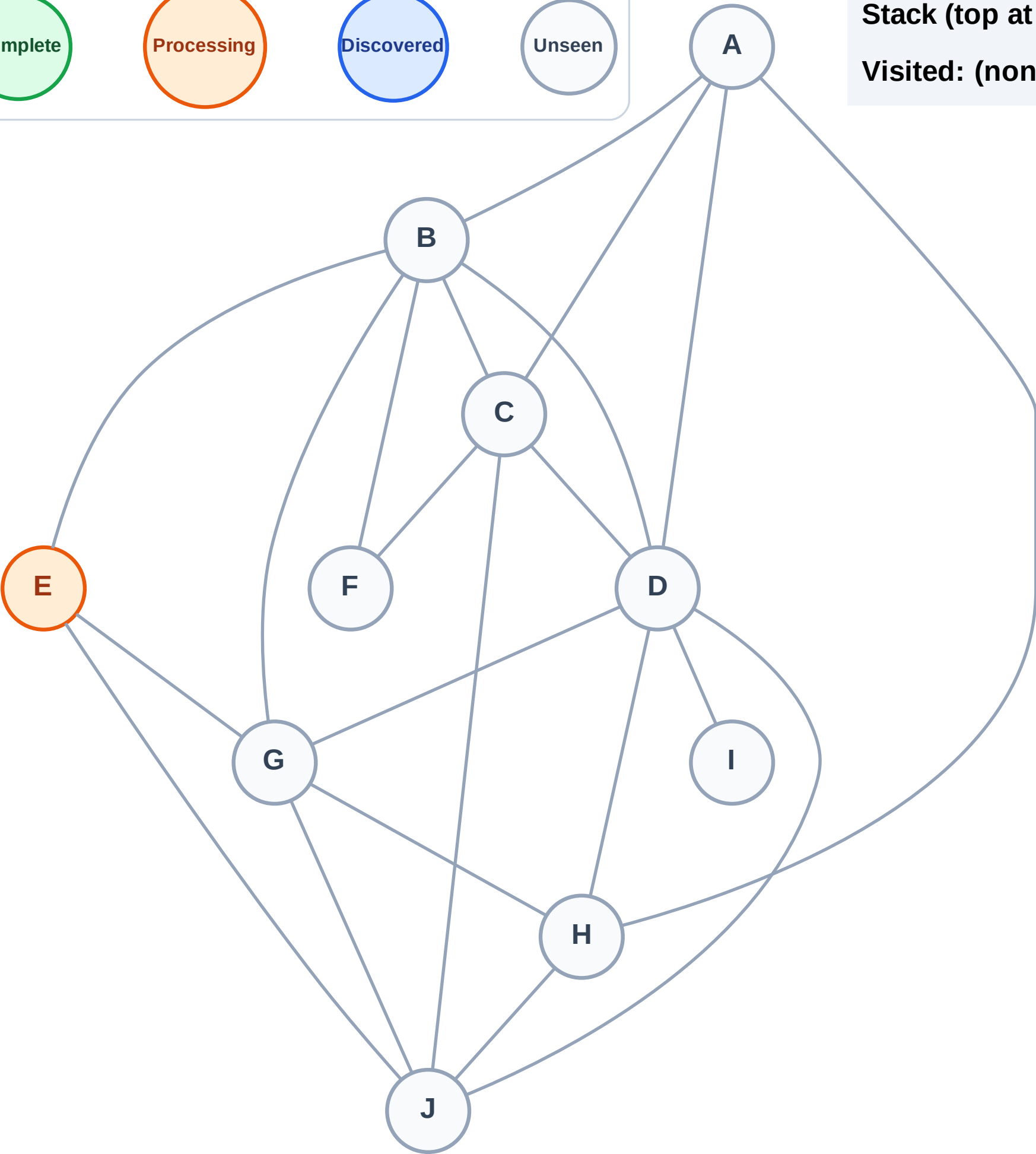
Complete

Processing

Discovered

Unseen

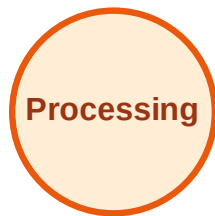
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

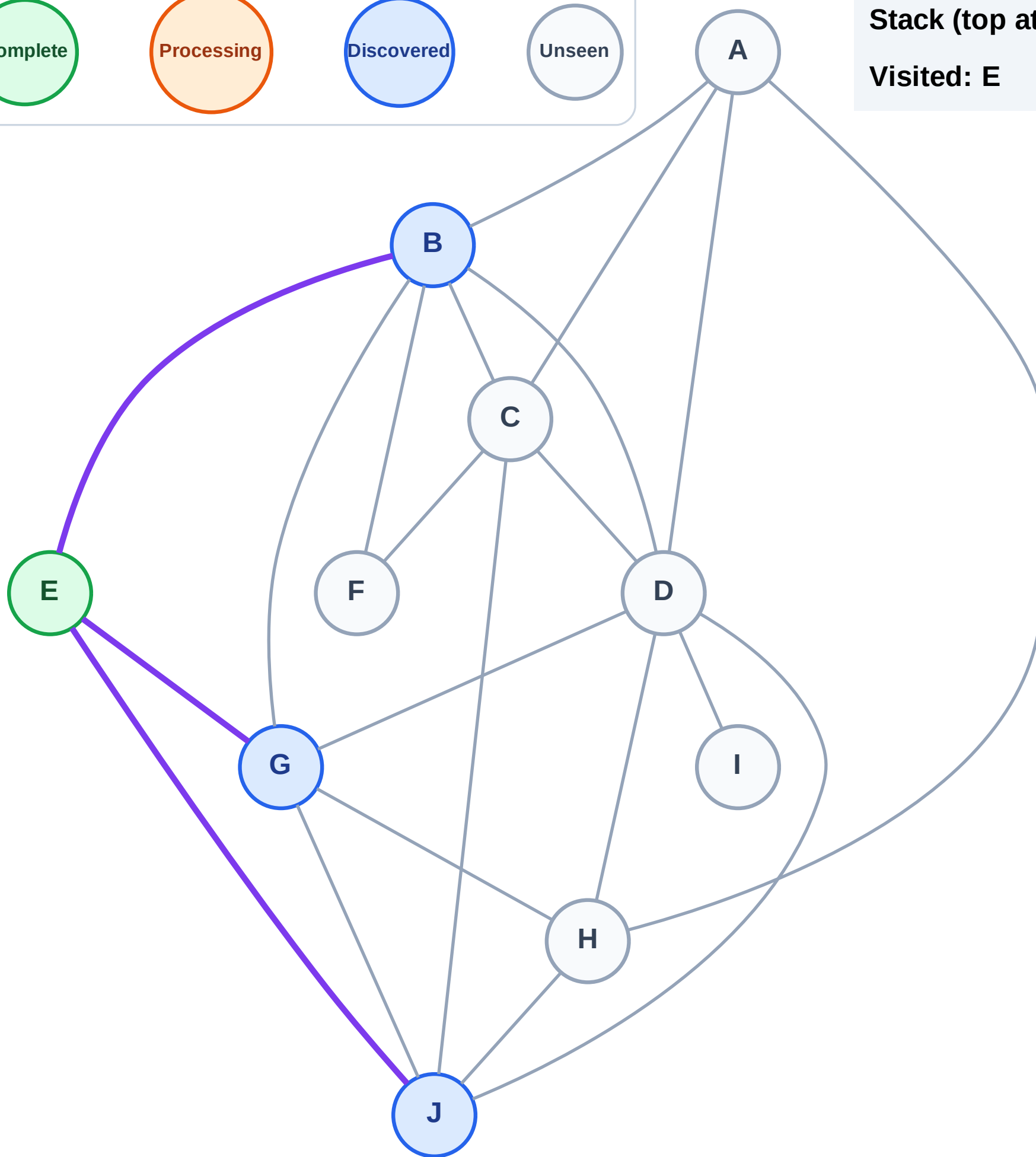
Step 3: Discover J, G, B from E

Legend



Stack (top at right): J | G | B

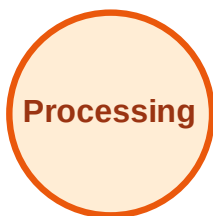
Visited: E



DFS Traversal

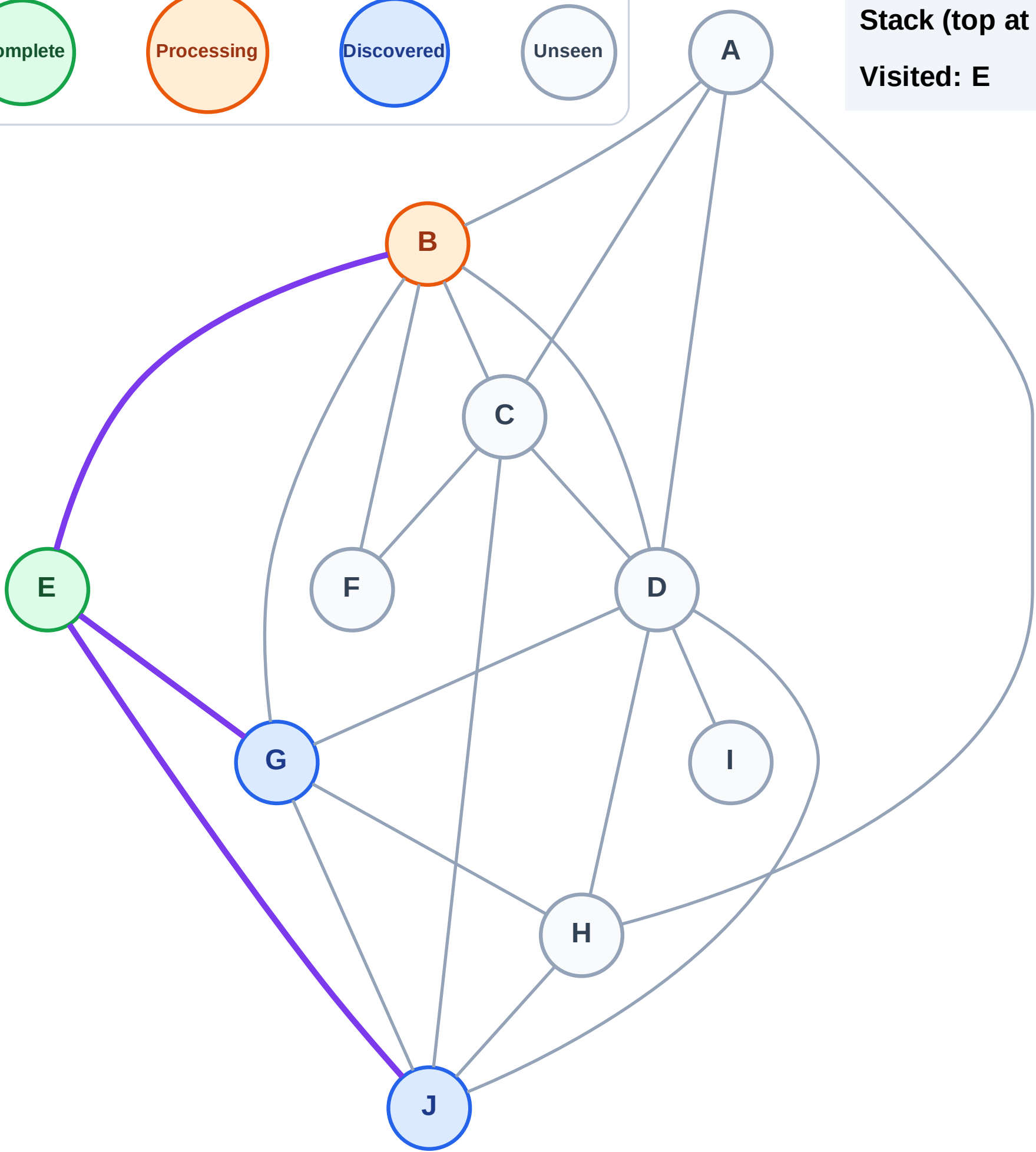
Step 4: Process node B

Legend



Stack (top at right): J | G

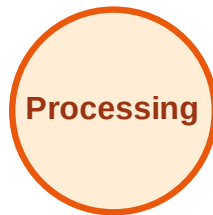
Visited: E



DFS Traversal

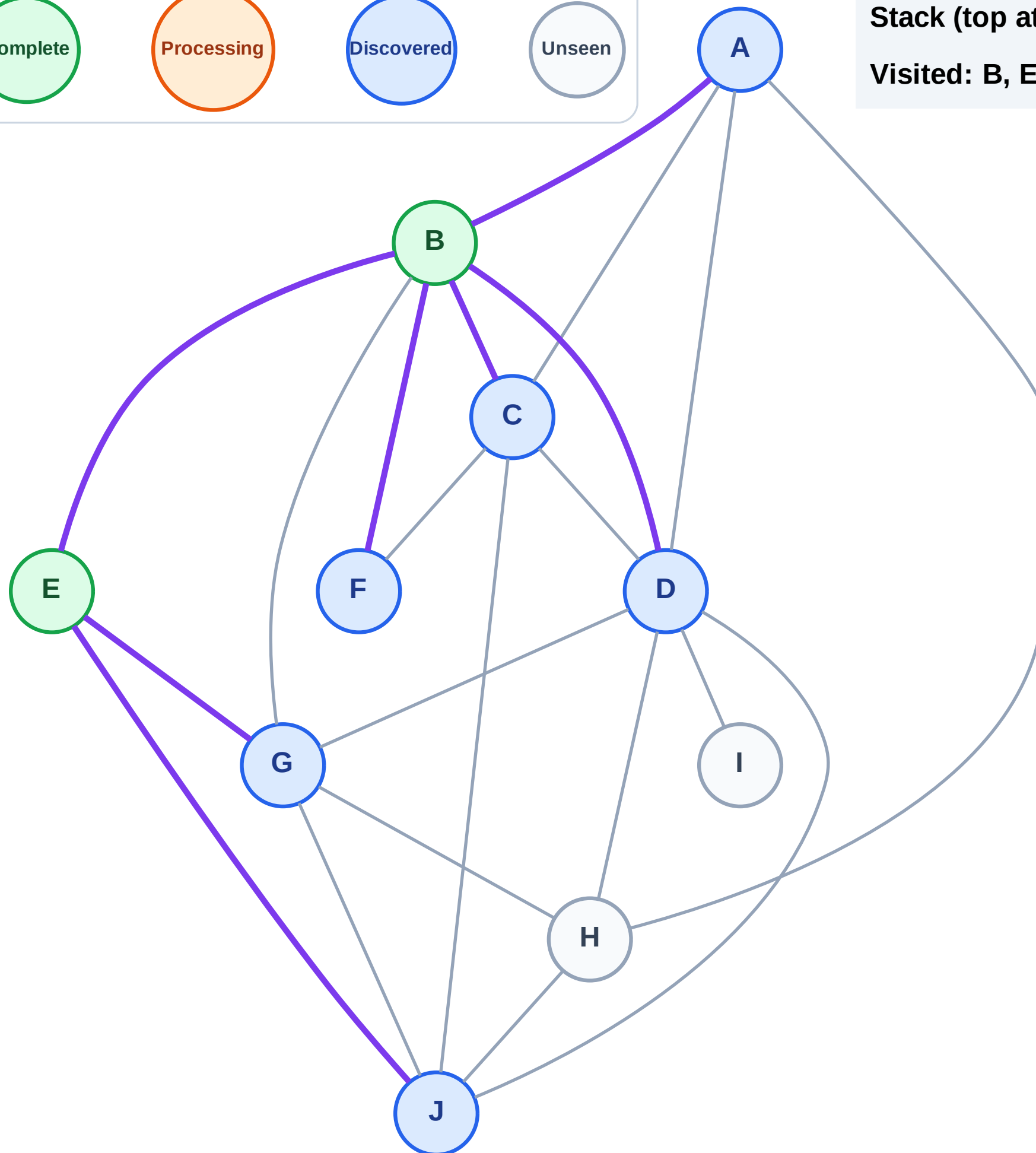
Step 5: Discover F, D, C, A from B

Legend



Stack (top at right): J | G | F | D | C | A

Visited: B, E



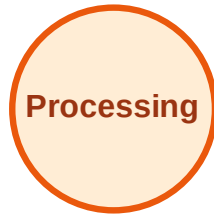
DFS Traversal

Step 6: Process node A

Legend



Complete



Processing



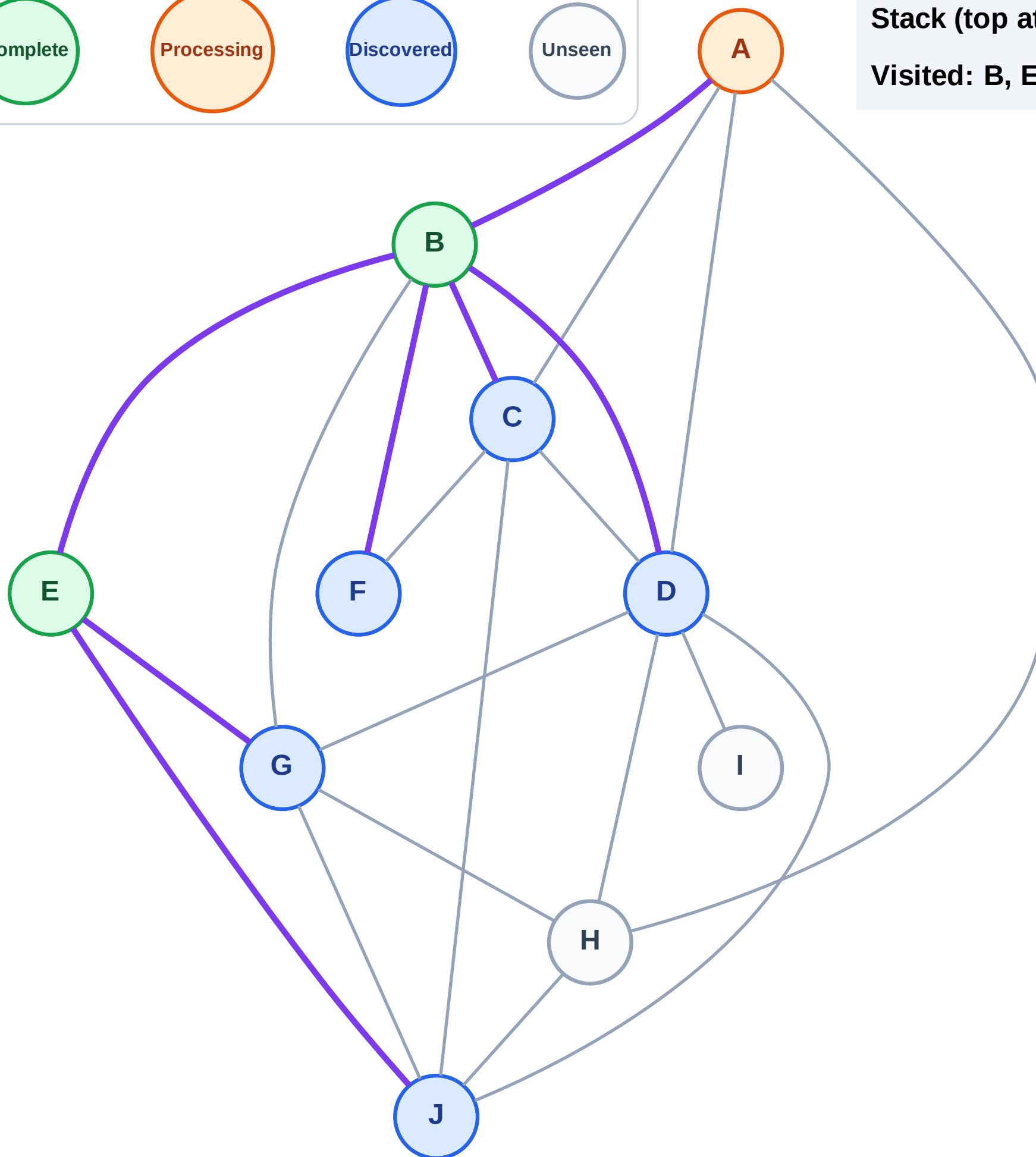
Discovered



Unseen

Stack (top at right): J | G | F | D | C

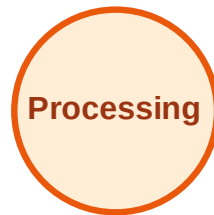
Visited: B, E



DFS Traversal

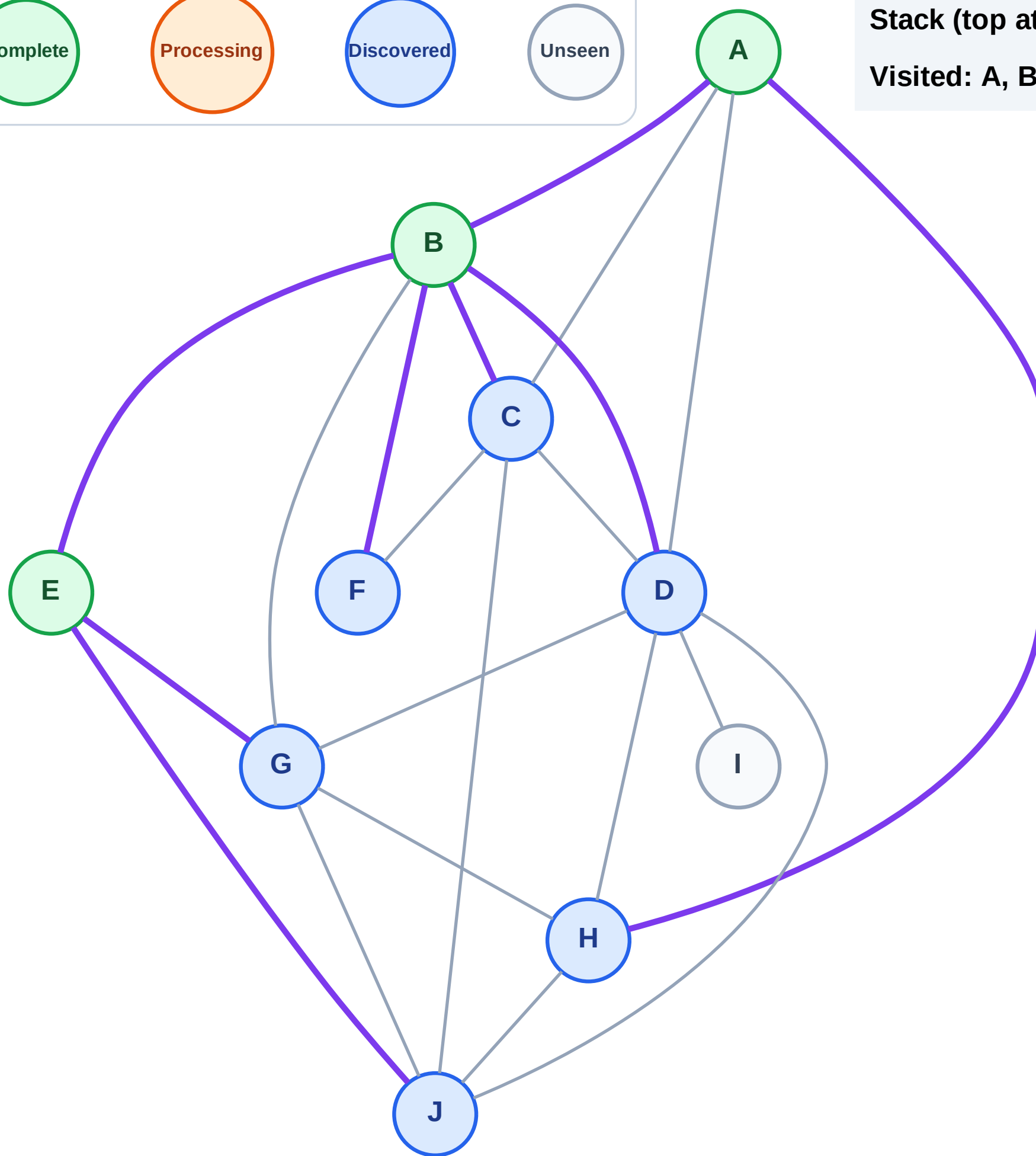
Step 7: Discover H from A

Legend



Stack (top at right): J | G | F | D | C | H

Visited: A, B, E



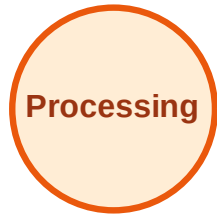
DFS Traversal

Step 8: Process node H

Legend



Complete



Processing



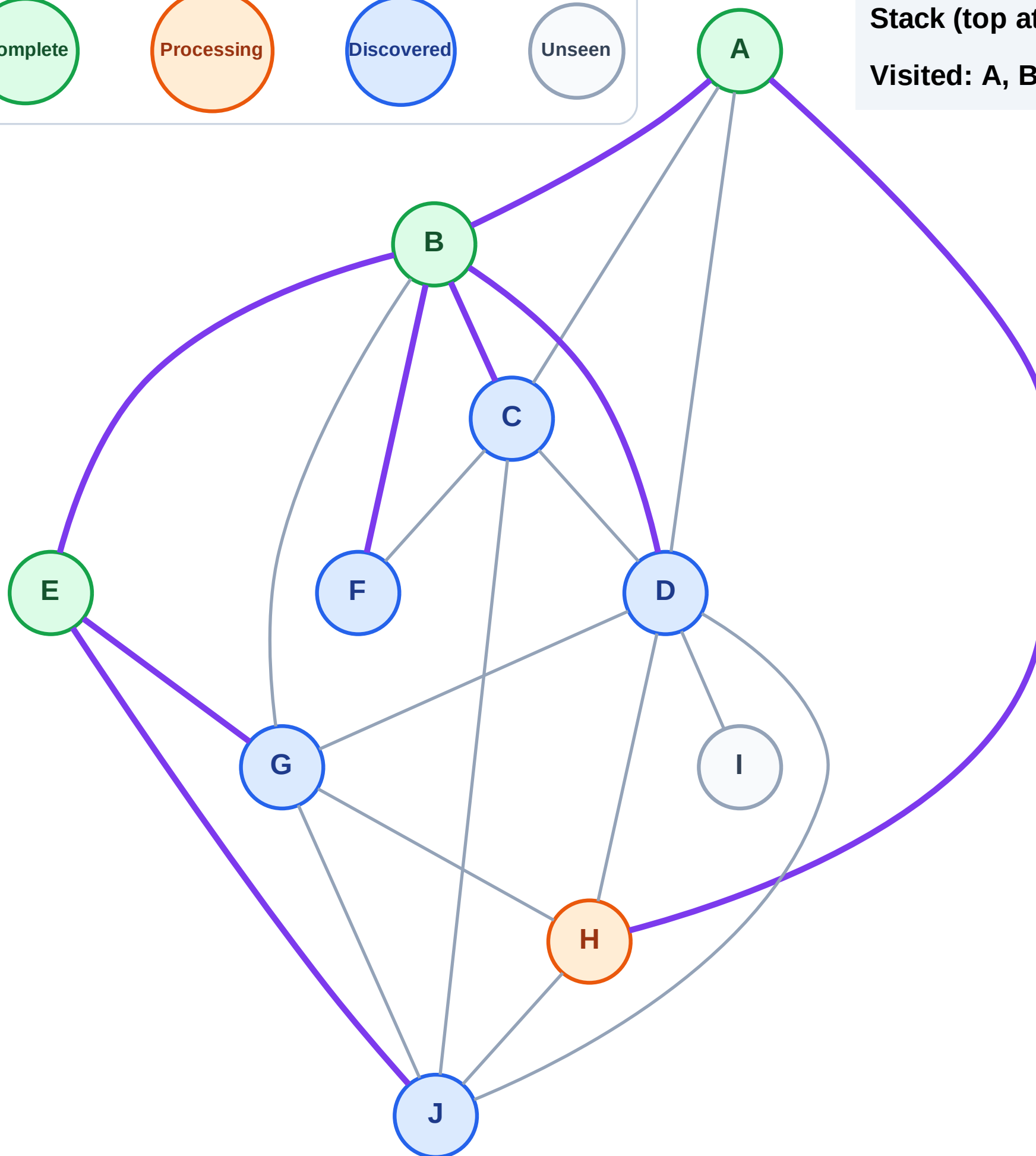
Discovered



Unseen

Stack (top at right): J | G | F | D | C

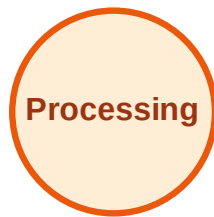
Visited: A, B, E



DFS Traversal

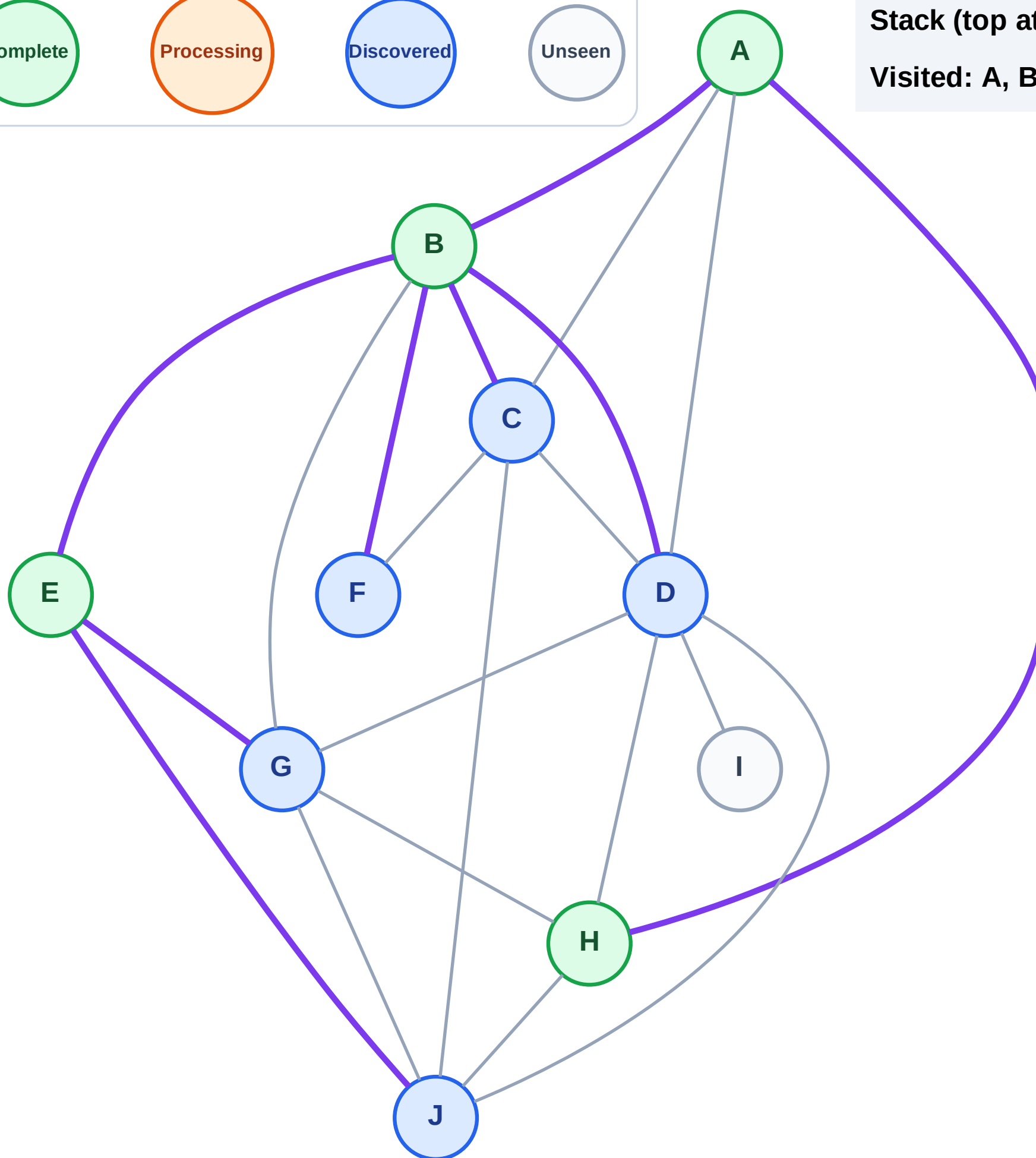
Step 9: No new neighbors from H

Legend



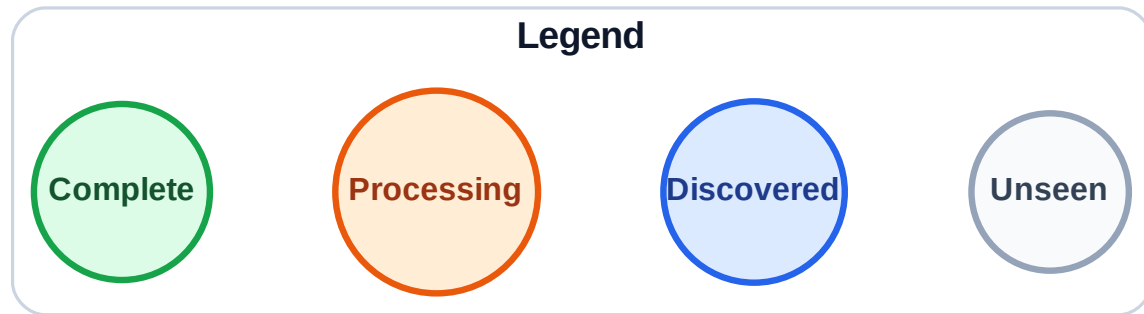
Stack (top at right): J | G | F | D | C

Visited: A, B, E, H



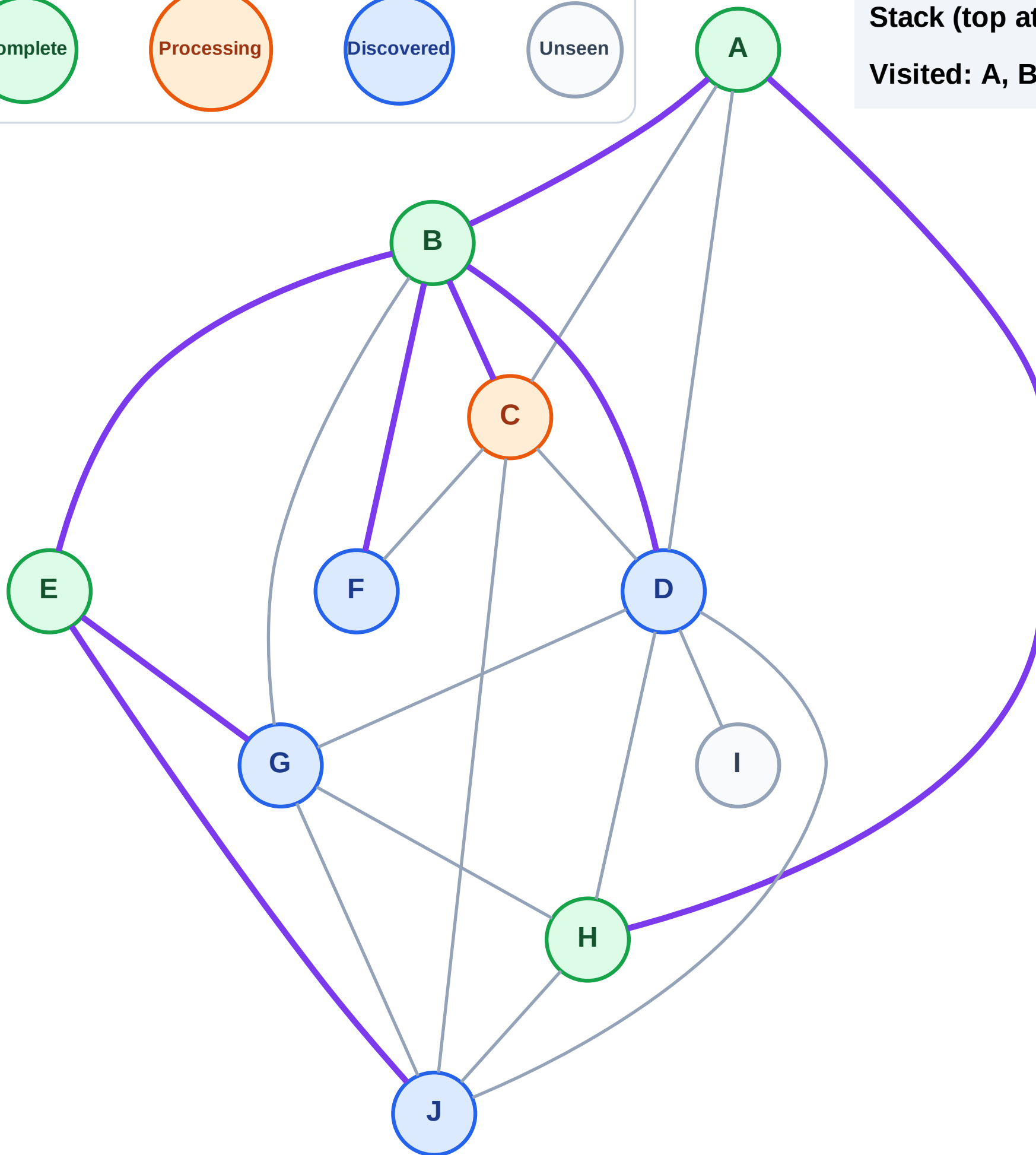
DFS Traversal

Step 10: Process node C



Stack (top at right): J | G | F | D

Visited: A, B, E, H



DFS Traversal

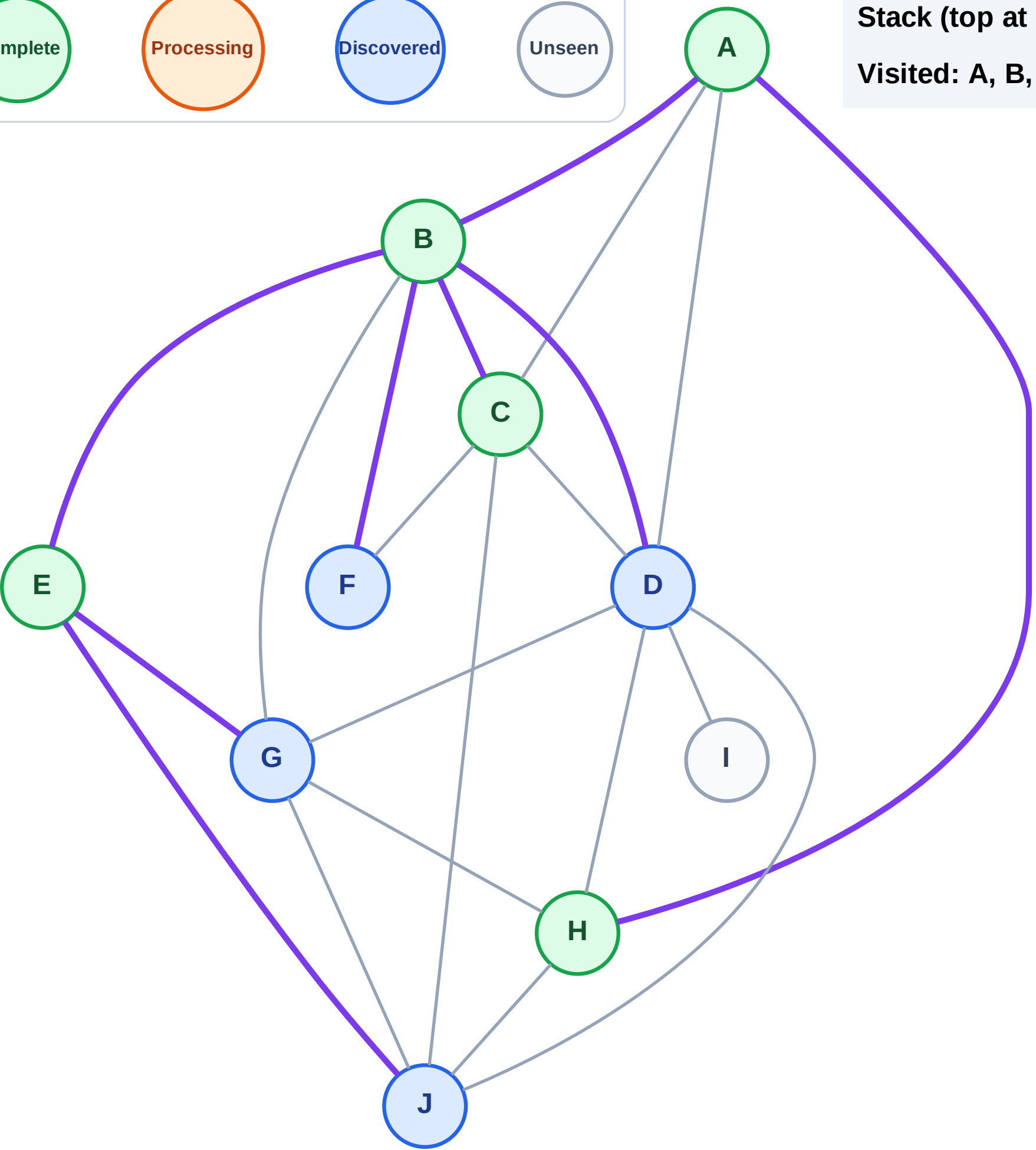
Step 11: No new neighbors from C

Legend



Stack (top at right): J | G | F | D

Visited: A, B, C, E, H



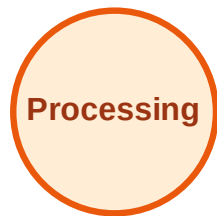
DFS Traversal

Step 12: Process node D

Legend



Complete



Processing



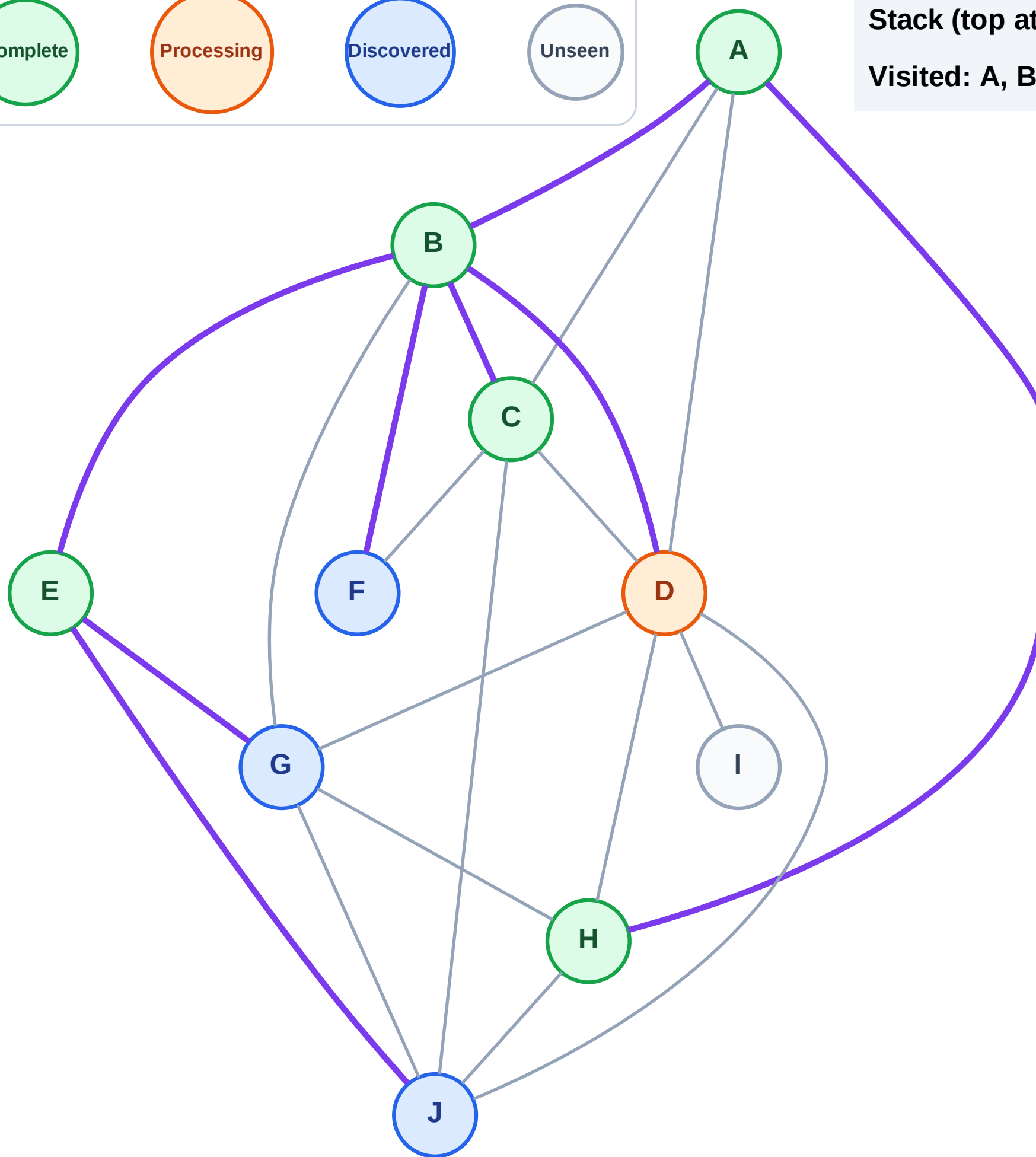
Discovered



Unseen

Stack (top at right): J | G | F

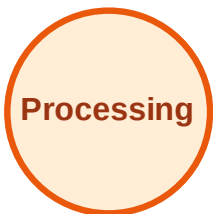
Visited: A, B, C, E, H



DFS Traversal

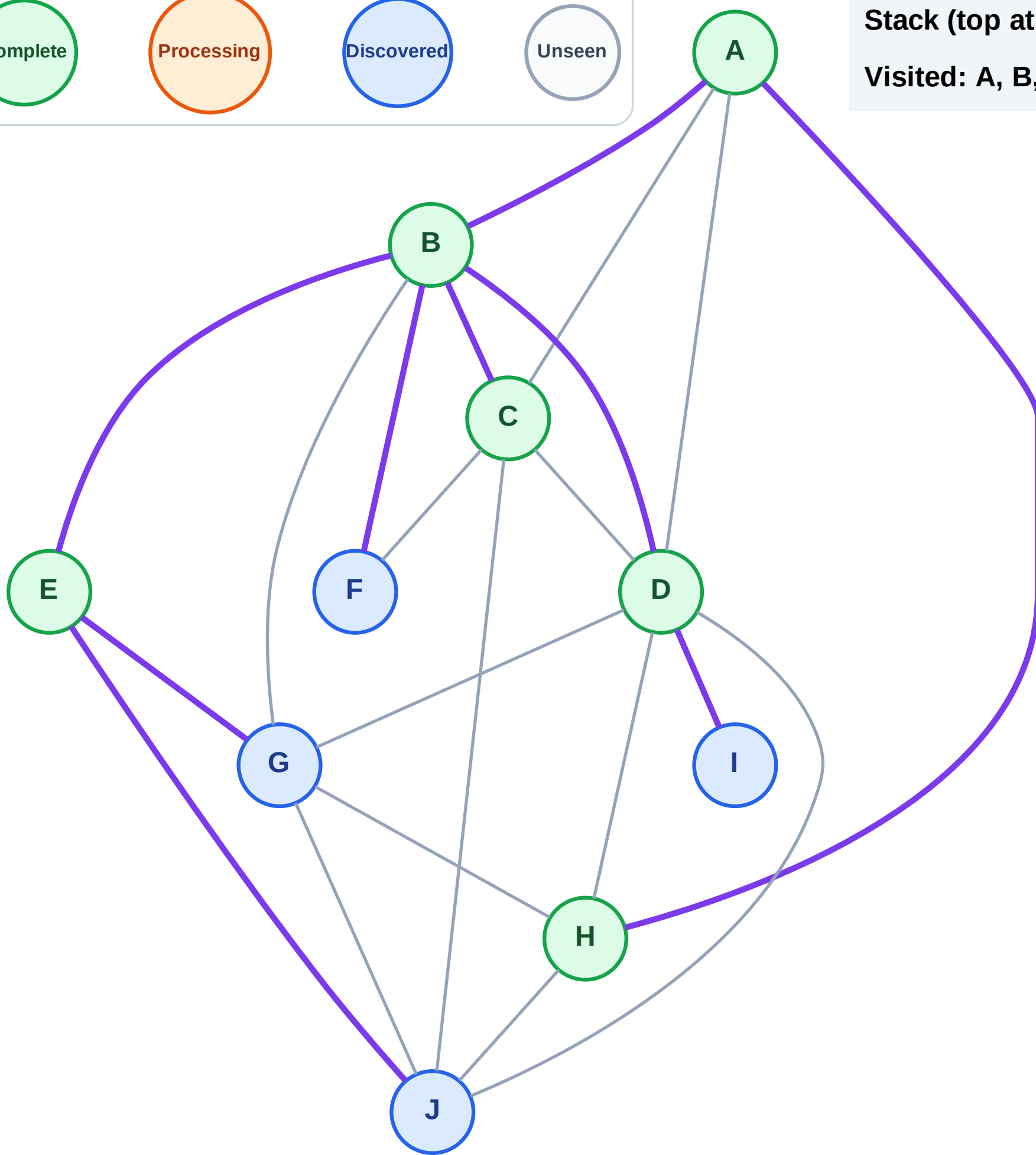
Step 13: Discover I from D

Legend



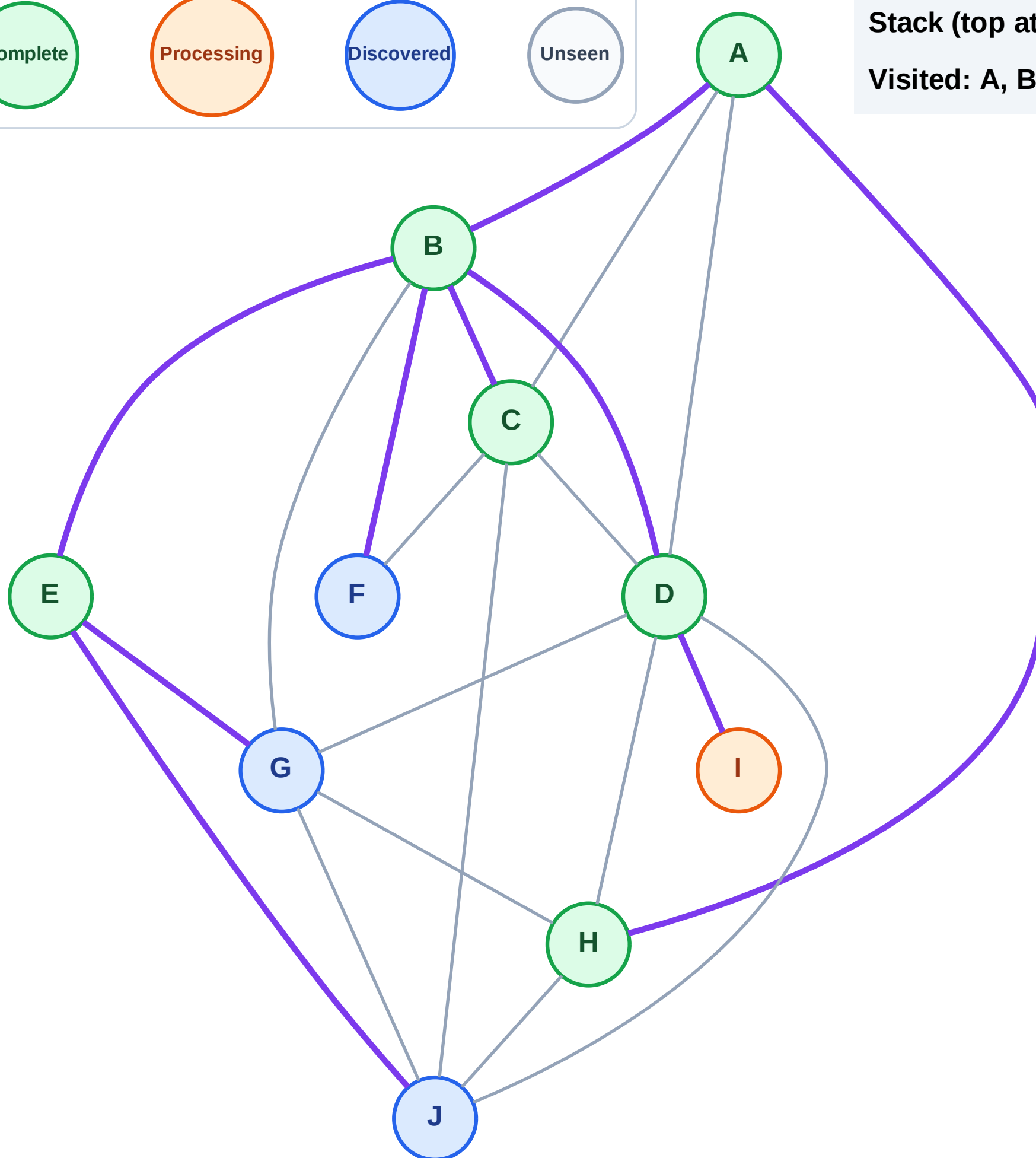
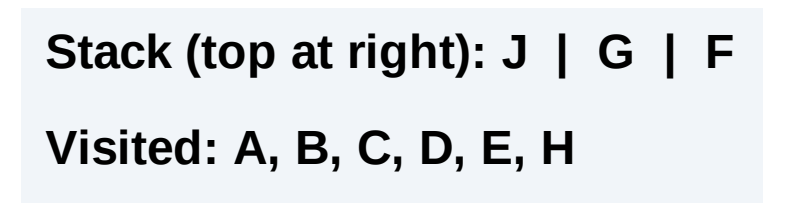
Stack (top at right): J | G | F | I

Visited: A, B, C, D, E, H



Step 14: Process node I

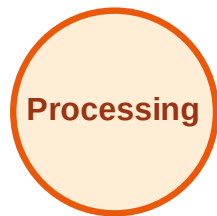
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DFS Traversal

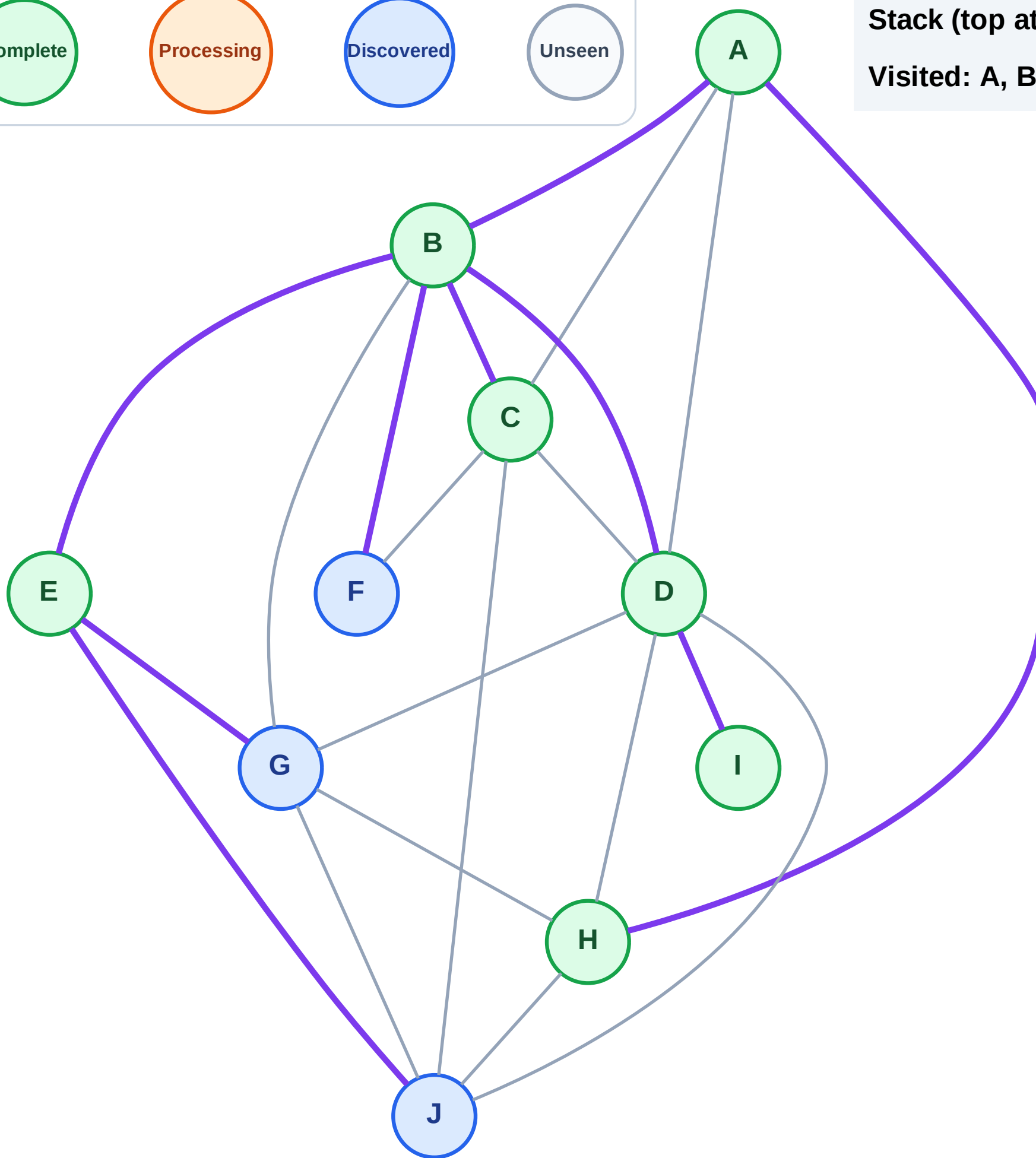
Step 15: No new neighbors from I

Legend



Stack (top at right): J | G | F

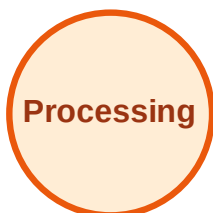
Visited: A, B, C, D, E, H, I



DFS Traversal

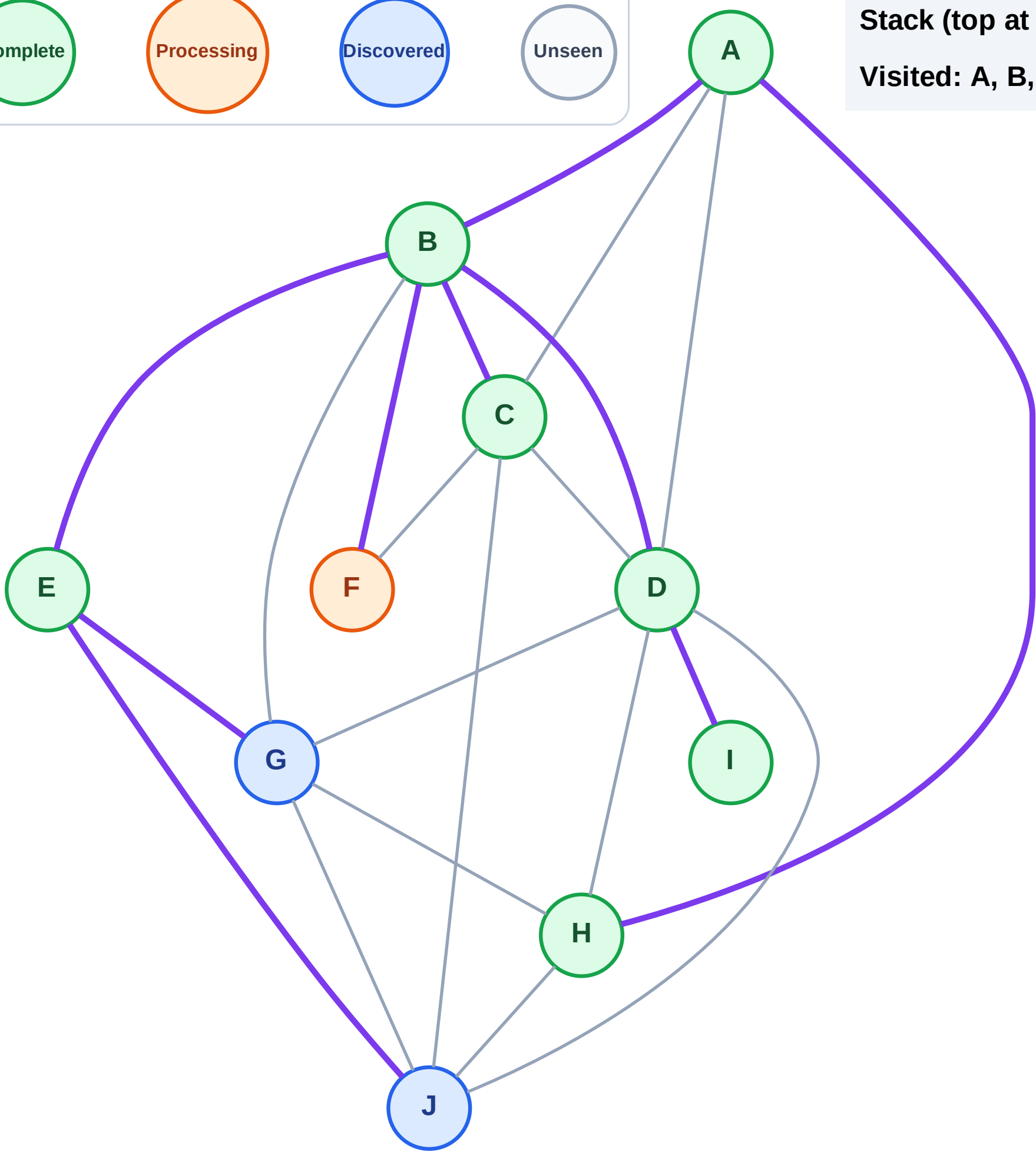
Step 16: Process node F

Legend



Stack (top at right): J | G

Visited: A, B, C, D, E, H, I



DFS Traversal

Step 17: No new neighbors from F

Legend

Complete

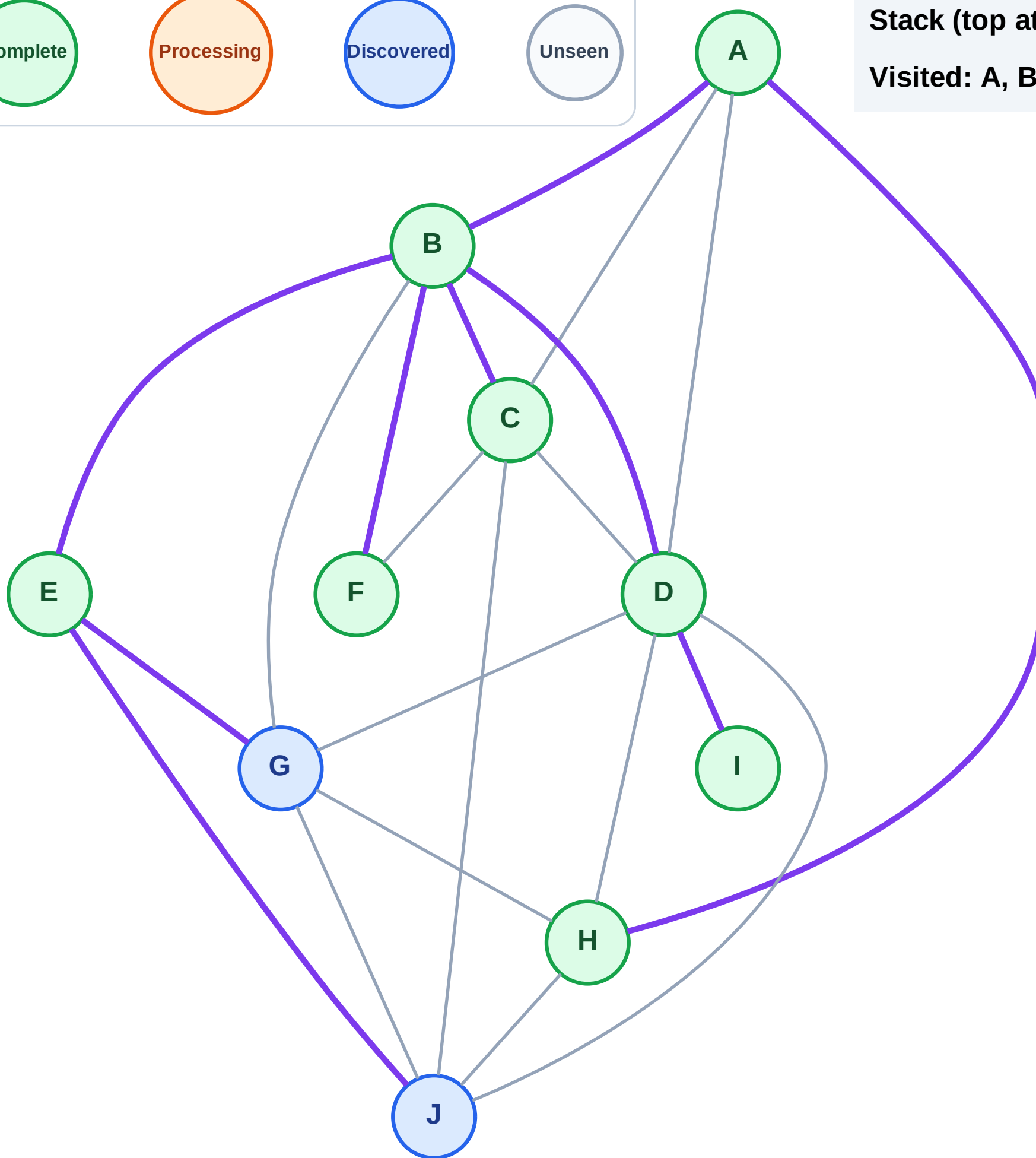
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node G

Legend

Complete

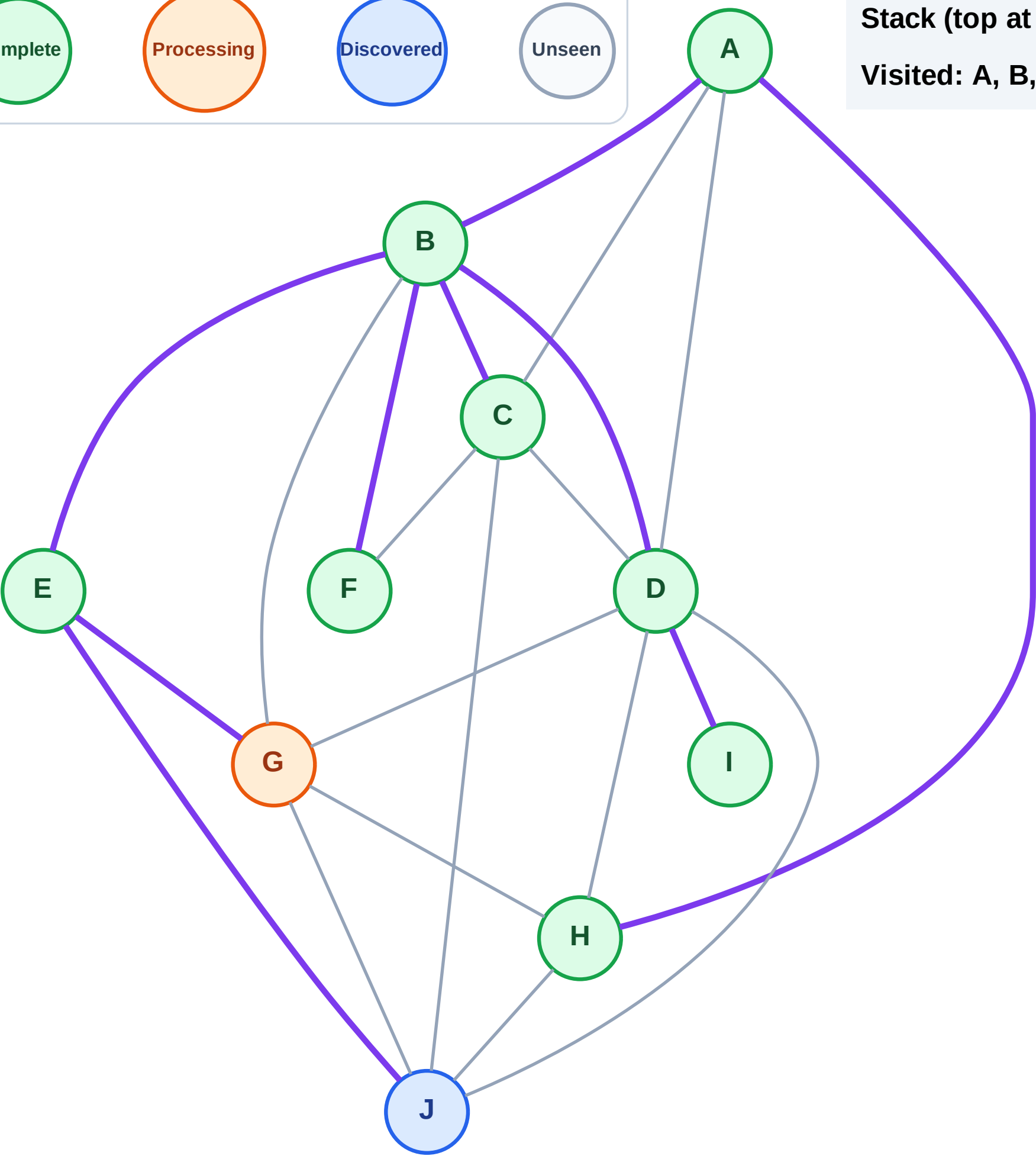
Processing

Discovered

Unseen

Stack (top at right): J

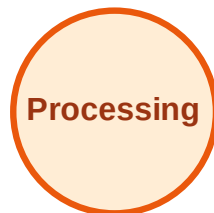
Visited: A, B, C, D, E, F, H, I



DFS Traversal

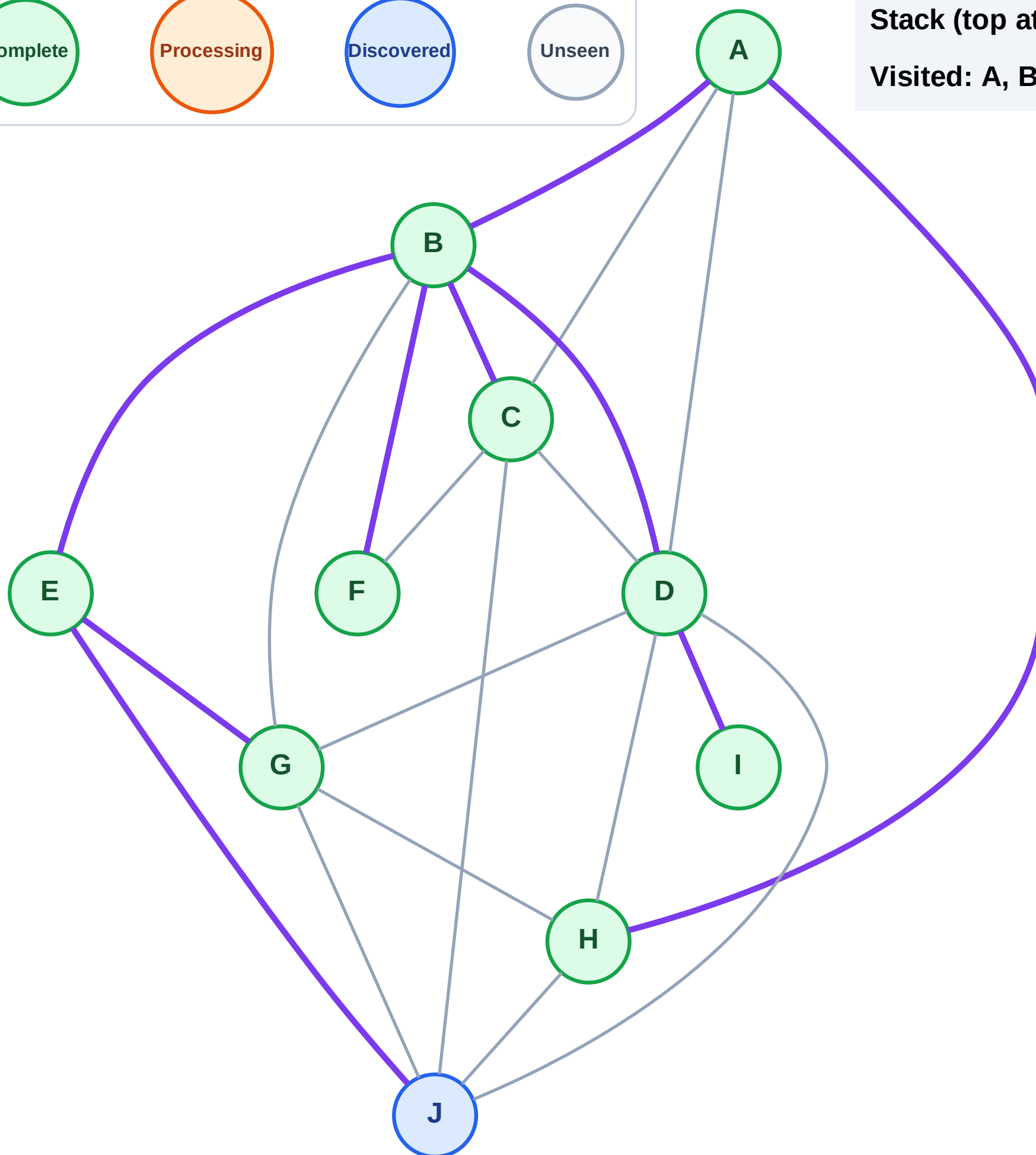
Step 19: No new neighbors from G

Legend



Stack (top at right): J

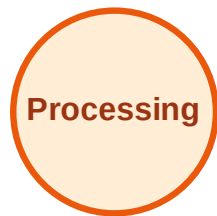
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

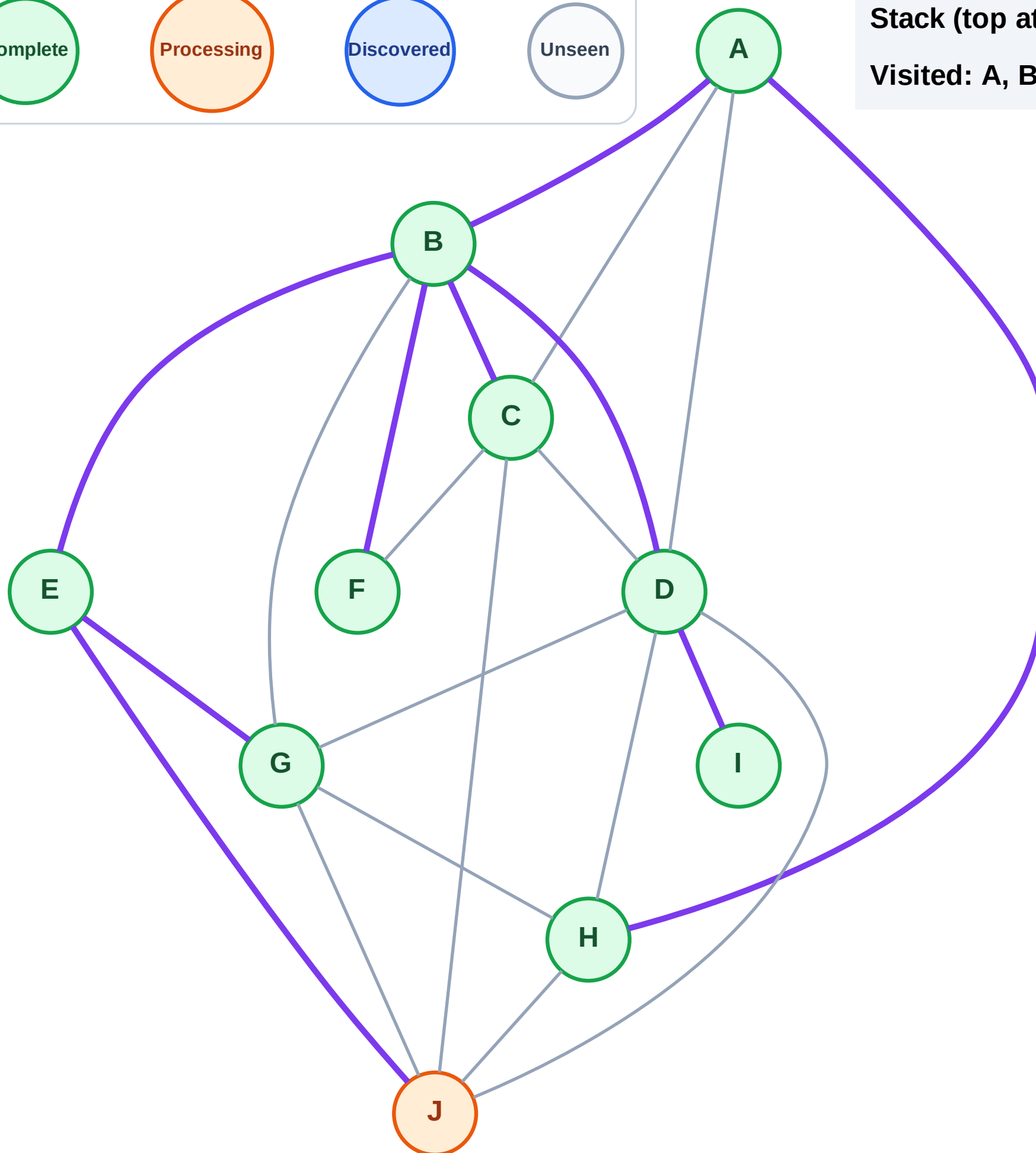
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

